



Northern University of Bussiness and Technology Khulna

NUBTK $\$(\hat{w})\$$

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DUET IUPC

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- 2 Mathematics
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Contest (1)

739397.cpp19 lines

```
#include <bits/stdc++.h>
using namespace std;

template<class T> using V = vector<T>;
typedef long long ll;
typedef pair<int, int> pii;

#define f first
#define s second
#define sz(x) (int)(x).size()
#define all(x) begin(x), end(x)
#define each(a, x) for(auto& a : x)
#define FOR(i, a, b) for(int i = (a); i < (b); ++i)
#define dbg(x) cerr <<__LINE__<<"| "<<#x<<" = "<<x<<endl

int main() {
    cin.sync_with_stdio(0); cin.tie(0);
    cin.exceptions(cin.failbit);
}

run.sh13 lines

set -e

# optional color
C='\033[0;31m'
R='\033[0m'

g++ -std=c++23 -O2 -Wall -Wextra -Wshadow -Wconversion \
-fsanitize=address -fsanitize=undefined \
-fsanitize=signed-integer-overflow -o $1 $1.cpp;

./$1 < input.txt 2> error.txt

echo -e "${C}$(cat error.txt){R}"
```

keybind.json9 lines

```
[
  {
    "key": "alt+r",
    "command": "workbench.action.terminal.sendSequence",
    "args": {
      "text": "bash run.sh ${fileBasenameNoExtension}\n"
    }
  }
]
```

hash.sh3 lines

```
# Hash file ignoring whitespace and comments. Verifies that
# code was correctly typed. Usage: 'sh hash.sh < A.cpp'
cpp -dD -P -fpreprocessed|tr -d '[:space:]'|md5sum|cut -c-6
```

gen.cpp173 lines

```
////////// DISTRIBUTIONS
mt19937 rng(chrono::steady_clock::now().time_since_epoch()).
    ↪count();

// return int in [L,R] inclusive
int rng_int(int L, int R) { assert(L <= R);
    return uniform_int_distribution<int>(L, R)(rng);
}
int64_t rng_int64(int64_t L, int64_t R) { assert(L <= R);
    return uniform_int_distribution<int64_t>(L, R)(rng);
}

// return double in [L,R] inclusive
double rng_db(double L, double R) { assert(L <= R);
    return uniform_real_distribution<double>(L, R)(rng);
}

// http://cplusplus.com/reference/random/geometric_distribution
    ↪geometric_distribution/
// flip a coin which is heads with probability p until you flip
    ↪heads
// mean value of c is 1/p-1
int rng_geo(double p) { assert(0 < p && p <= 1); // p large ->
    ↪closer to 0
    return geometric_distribution<int>(p)(rng);
}

////////// VECTORS + PERMS

// shuffle a vector
template<class T> void randShuffle(vector<T>& v) {
    shuffle(begin(v), end(v), rng);
}

// generate random permutation of [0,N-1]
vector<int> randPermutation(int N) {
    vector<int> v(N); iota(begin(v), end(v), 0);
    randShuffle(v); return v;
}

// random permutation of [0,N-1] with first element 0
vector<int> randPermutationZero(int N) {
    vector<int> v(N-1); iota(begin(v), end(v), 1);
    randShuffle(v);
    v.insert(begin(v), 0); return v;
}

// shuffle permutation of [0,N-1]
vector<int> shufPerm(vector<int> v) {
    int N = size(v);
```

```
vector<int> key = randPermutation(N);
vector<int> res(N);
for(int i = 0; i < N; i++) res[key[i]] = key[v[i]];
return res;
}

// vector with all entries in [L,R]
vector<int> randVector(int N, int L, int R) {
    vector<int> v(N);
    for (int i = 0; i < N; i++) {
        v[i] = rng_int(L, R);
    }
    return v;
}

// vector with all entries in [L,R], unique
vector<int> rngDistinctVector(int N, int L, int R) {
    set<int> used; vector<int> v;
    while ((int)size(v) < N) {
        int x = rng_int(L, R);
        if (!used.count(x)) {
            used.insert(x);
            v.push_back(x);
        }
    } return v;
}

// generate random prime in [L,R]
int randPrime(int L, int R) { assert(L <= R && L >= 2);
    while(1) {
        int x = rng_int(L, R);
        bool bad = 0;
        for (int i = 2; i*i <= x; i++) if (x%i == 0) bad = 1;
        if (!bad) return x;
    }
}

////////// GRAPHS

// relabel edges ed according to perm, shuffle
vpi relabelAndShuffle(vpi ed, vi perm) {
    each(t,ed) {
        t.f = perm[t.f], t.s = perm[t.s];
        if (rng()&1) swap(t.f,t.s);
    }
    shuf(ed); return ed;
}

// shuffle graph with vertices [0,N-1]
vpi shufGraph(int N, vpi ed) { // randomly swap endpoints,
    ↪rearrange labels
    return relabelAndShuffle(ed,randPerm(N)); }
vpi shufGraphZero(int N, vpi ed) {
    return relabelAndShuffle(ed,randPermZero(N)); }

// shuffle tree given N-1 edges
vpi shufTree(vpi ed) { return shufGraph(sz(ed)+1,ed); }
// randomly swap endpoints, rearrange labels
vpi shufRootedTree(vpi ed) {
    return relabelAndShuffle(ed,randPermZero(sz(ed)+1)); }

void pgraphOne(int N, vpi ed) {
    ps(N,sz(ed));
    each(e,ed) ps(1+e.f,1+e.s);
}

////////// GENERATING TREES

// for generating tall tree
```

```
pi geoEdge(int i, db p) { assert(i > 0);
    return {i,max(0,i-1-rng_geo(p))}; }

// generate edges of tree with verts [0,N-1]
// smaller back -> taller tree
vpi treeRand(int N, int back) {
    assert(N >= 1 && back >= 0); vpi ed;
    FOR(i,1,N) ed.eb(i,i-1-rng_int(0,min(back,i-1)));
    return ed; }

// generate path
vpi path(int N) { return treeRand(N,0); }

// generate tall tree (large diameter)
// the higher the p the taller the tree
vpi treeTall(int N, db p) { assert(N >= 1);
    vpi ed; FOR(i,1,N) ed.pb(geoEdge(i,p));
    return ed; }

// generate tall tree, then add rand at end
vpi treeTallShort(int N, db p) {
    assert(N >= 1); int mid = (N+1)/2;
    vpi ed = treeTall(mid,p);
    FOR(i,mid,N) ed.eb(i,rng_int(0,i-1));
    return ed; }

// lots of stuff connected to either heavy1 or heavy2
vpi treeTallHeavy(int N, db p) {
    assert(N >= 1); // + bunch of rand
    vpi ed; int heavy1 = 0, heavy2 = N/2;
    FOR(i,1,N) {
        if(i < N/4) ed.eb(i,heavy1);
        else if (i > heavy2 && i < 3*N/4) ed.eb(i,heavy2);
        else ed.pb(geoEdge(i,p));
    }
    return ed;
}

// heavy tall tree + random
// lots of verts connected to heavy1 or heavy2
vpi treeTallHeavyShort(int N, db p) {
    assert(N >= 1); // + almost-path + rand
    vpi ed; int heavy1 = 0, heavy2 = N/2;
    FOR(i,1,N) {
        if(i < N/4) ed.eb(i,heavy1);
        else if (i <= heavy2) ed.pb(geoEdge(i,p)); // tall ->
            ↪heavy1
        else if (i > heavy2 && i < 3*N/4) ed.eb(i,heavy2);
        else ed.eb(i,rng_int(0,i-1));
    }
    return ed;
}

int main(int, char* argv[]) {
    int tc; sscanf(argv[1],"%d",&tc); // test case #
    rng.seed(tc);
    // sscanf(argv[2],"%d",&b); // random seed
    // sscanf(argv[3],"%d",&c); // also random seed
}
```

stress.sh10 lines

```
# A and B are executables you want to compare, gen takes int
# as command line arg. Usage: 'sh stress.sh'
for((i = 1; ; ++i)); do
    echo $i
    ./gen $i > in
    ./sol < in > outSol
```

```
./slow < in > outSlow
diff -w outSol outSlow || break
# diff -w <(. /A < in) <(. /B < in) || break
done
```

troubleshoot.txt75 lines

General:
Write down most of your thoughts, even if you’re not sure whether they’re useful.
Give your variables (and files) meaningful names.
Stay organized and don’t leave papers all over the place!
You should know what your code is doing ...

Pre-submit:
Write a few simple test cases if sample is not enough.
Are time limits close? If so, generate max cases.
Is the memory usage fine?
Could anything overflow?
Remove debug output.
Make sure to submit the right file.

Wrong answer:
Print your solution! Print debug output as well.
Read the full problem statement again.
Have you understood the problem correctly?
Are you sure your algorithm works?
Try writing a slow (but correct) solution.
Can your algorithm handle the whole range of input?
Did you consider corner cases (ex. n=1)?
Is your output format correct? (including whitespace)
Are you clearing all data structures between test cases?
Any uninitialized variables?
Any undefined behavior (array out of bounds)?
Any overflows or NaNs (or shifting ll by >=64 bits)?
Confusing N and M, i and j, etc.?
Confusing ++i and i++?
Return vs continue vs break?
Are you sure the STL functions you use work as you think?
Add some assertions, maybe resubmit.
Create some test cases to run your algorithm on.
Go through the algorithm for a simple case.
Go through this list again.
Explain your algorithm to a teammate.
Ask the teammate to look at your code.
Go for a small walk, e.g. to the toilet.
Rewrite your solution from the start or let a teammate do it.

Geometry:
Work with ints if possible.
Correctly account for numbers close to (but not) zero. Related:
for functions like acos make sure absolute val of input is not (slightly) greater than one.
Correctly deal with vertices that are collinear, concyclic, coplanar (in 3D), etc.
Subtracting a point from every other (but not itself)?

Runtime error:
Have you tested all corner cases locally?
Any uninitialized variables?
Are you reading or writing outside the range of any vector?
Any assertions that might fail?
Any possible division by 0? (mod 0 for example)
Any possible infinite recursion?
Invalidated pointers or iterators?
Are you using too much memory?
Debug with resubmits (e.g. remapped signals, see Various).

Time limit exceeded:

Do you have any possible infinite loops?
What’s your complexity? Large TL does not mean that something simple (like NlogN) isn’t intended.
Are you copying a lot of unnecessary data? (References)
Avoid vector, map. (use arrays/unordered_map)
How big is the input and output? (consider FastIO)
What do your teammates think about your algorithm?
Calling count() on multiset?

Memory limit exceeded:
What is the max amount of memory your algorithm should need?
Are you clearing all data structures between test cases?
If using pointers try BumpAllocator.

Mathematics (2)

2.1 Trigonometry

$$\sin(v + w) = \sin v \cos w + \cos v \sin w$$

$$\cos(v + w) = \cos v \cos w - \sin v \sin w$$

$$\tan(v + w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$

$$\sin v + \sin w = 2 \sin \frac{v + w}{2} \cos \frac{v - w}{2}$$

$$\cos v + \cos w = 2 \cos \frac{v + w}{2} \cos \frac{v - w}{2}$$

$$a \cos x + b \sin x = r \cos(x - \phi)$$

$$a \sin x + b \cos x = r \sin(x + \phi)$$

where $r = \sqrt{a^2 + b^2}, \phi = \operatorname{atan2}(b, a)$.

2.2 Geometry

2.2.1 Triangles

Side lengths: a, b, c

Semiperimeter: $s = \frac{a + b + c}{2}$

Area: $A = \sqrt{s(s - a)(s - b)(s - c)}$

Circumradius: $R = \frac{abc}{4A}$

Inradius: $r = \frac{A}{p}$

Length of median (divides triangle into two equal-area triangles):
 $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Length of bisector (divides angles in two):

$$s_a = \sqrt{bc \left[1 - \left(\frac{a}{b + c} \right)^2 \right]}$$

Law of sines: $\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R}$

Law of cosines: $a^2 = b^2 + c^2 - 2bc \cos \alpha$

Law of tangents: $\frac{a + b}{a - b} = \frac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}}$

2.2.2 Quadrilaterals

With side lengths a, b, c, d , diagonals e, f , diagonals angle θ , area A and magic flux $F = b^2 + d^2 - a^2 - c^2$:

$$4A = 2ef \cdot \sin \theta = F \tan \theta = \sqrt{4e^2 f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is 180° , $ef = ac + bd$, and $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$.

2.3 Derivatives/Integrals

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx} \tan x = 1 + \tan^2 x$$
$$\int \tan ax = -\frac{\ln|\cos ax|}{a}$$
$$\int e^{-x^2} = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x)$$

$$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$
$$\int x \sin ax = \frac{\sin ax - ax \cos ax}{a^2}$$
$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1)$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

2.4 Sums

$$r^a + r^{a+1} + \cdots + r^b = \frac{r^{b+1} - r^a}{r - 1}, r \neq 1$$
$$1 + r + r^2 + \cdots + r^\infty = \frac{1}{1 - r}, \quad |r| < 1$$

$$a+(a+d)r+(a+2d)r^2+(a+3d)r^3+\cdots = \frac{a}{1-r} + \frac{dr}{(1-r)^2}, \quad |r| < 1$$
$$a+(a+d)+(a+2d)+\cdots+(a+(n-1)d) = \frac{n}{2}(2a+(n-1)d)$$

$$1+2+3+\cdots+n = \frac{n(n+1)}{2}$$
$$1+3+5+\cdots+2n-1 = n^2$$
$$2+4+6+\cdots+2n = n(n+1)$$

$$1^2+2^2+3^2+\cdots+n^2 = \frac{n(2n+1)(n+1)}{6}$$
$$1^3+2^3+3^3+\cdots+n^3 = \frac{n^2(n+1)^2}{4}$$
$$1^4+2^4+3^4+\cdots+n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

2.5 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, (-\infty < x < \infty)$$
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (-1 < x \leq 1)$$
$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \dots, (-1 \leq x \leq 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, (-\infty < x < \infty)$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, (-\infty < x < \infty)$$

2.6 Probability theory

Let X be a discrete random variable with probability $p_X(x)$ of assuming the value x . It will then have an expected value (mean) $\mu = \mathbb{E}(X) = \sum_x x p_X(x)$ and variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$ where σ is the standard deviation.

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

If X, Y are independent,

$$V(aX + bY) = a^2V(X) + b^2V(Y).$$

Expectation is linear If X is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.

Data Structures (3)

3.1 STL

MapComparator.h

Description: example of function object (functor) for map or set

Usage: set<int,cmp> s; map<int,int,cmp> m;

5bfa6c, 1 lines

struct cmp{bool operator()(int l,int r)const{return l>r;}};

HashMap.h

Description: Hash map with similar API as unordered_map. Initial capacity must be a power of 2 if provided.

Usage: gp_hash_table<int, int, chash> h({},{},{},{},{1<<16});

Memory: ~1.5x unordered map

Time: ~3x faster than unordered map

e58314, 12 lines

<ext/pb_ds/assoc.container.hpp>

using namespace __gnu_pbds;

struct chash {

const uint64_t C = 4e18*acos(0)+71;

const int R = chrono::steady_clock::now().time_since_epoch().count();

size_t operator()(uint64_t x) const {

return __builtin_bswap64((x^R)*C);

}

};

template <class K, class V> using HT = gp_hash_table<K, V,

<chash>;

template <class K, class V> V get(HT<K, V> &u, K x) {

auto it = u.find(x); return it == u.end() ? 0 : it->second;

}

OrderStatisticTree.h

Description: A set (not multiset!) with support for finding the n 'th element, and finding the index of an element. Change null_type to get a map.

Time: $\mathcal{O}(\log N)$

<ext/pb_ds/assoc.container.hpp>

222518, 18 lines

using namespace __gnu_pbds;

template<class T> using IndexedSet = tree<T, null_type,

less<T>, rb_tree_tag, tree_order_statistics_node_update>;

#define ook order_of_key // position of the given element

#define fbo find_by_order // return an iterator

// Number of element <= k

int atMost(IndexedSet<pair<int, int>>& T, int k) {

return T.order_of_key({k, INT_MAX});

}

// Number of element grater >= k

int atLeast(IndexedSet<pair<int, int>>& T, int k) {

return T.size() - T.order_of_key({k, INT_MIN});

}

int getSum(IndexedSet<pair<int, int>>& T, int l, int r) {

return atMost(T,r)-atMost(T,l-1);

}

CoordCompress.h

Description: coordinate compression, get index of x by counting number of element less than x

Time: $\mathcal{O}(N \log N)$

14ee69, 7 lines

template<class T> void compress(vector<T>& v) {

ranges::sort(v);

v.erase(unique(v.begin(), v.end()), v.end());

}

template<class T> int get(vector<T>& v, T x) {

return ranges::lower_bound(v, x) - begin(v);

}

SWAG.h

Description: SWAG (Sliding Window Aggregation) using two stacks. Supports queue operations with $\mathcal{O}(1)$ amortized aggregate queries. Works for any associative operation (OR, AND, XOR, GCD, MIN, MAX, SUM, etc).

Usage: Queue<int> q;

q.push(x);

q.pop();

q.query(); // aggregate over all elements in queue

Memory: $\mathcal{O}(n)$

Time: push $\mathcal{O}(1)$, pop $\mathcal{O}(1)$ amortized, query $\mathcal{O}(1)$

5aa600, 29 lines

template<class T> struct Stack {

vector<pair<T, T>> stk;

T cmb(T x) {

return stk.empty() ? x : min(stk.back().second, x);

}

void push(T x) { stk.push_back({x, cmb(x)}); }

void pop() { stk.pop_back(); }

T agg() const { return stk.back().second; }

};

template<class T> struct Queue {

Stack<T> in, out;

T cmb(T a, T b) {

return min(a, b);

}

void push(T x) { in.push(x); }

void pop() {

if (out.stk.empty()) while (!in.stk.empty()) {

out.push(in.stk.back().first);

in.pop();

}

}

```
        out.pop();
    }
    T query() {
    if (in.stk.empty()) return out.agg();
    if (out.stk.empty()) return in.agg();
    return cmb(in.agg(), out.agg());
    }
};
```

3.2 1D Range Queries

SparseTable.h

Description: Static 1D range (min/max/gcd/lcm/or/and) query. If TL is an issue, use arrays instead of vectors and store values instead of indices.
Memory: $\mathcal{O}(N \log N)$
Time: $\mathcal{O}(1)$

471d4b, 19 lines

```
template<class T> struct SparseTable {
    int N, LOG;
    vector<vector<T>> jmp;
    T cmb(T a, T b) { return min(a, b); }
    SparseTable(const vector<T>& v) {
        N = v.size(); LOG = __lg(N);
        jmp.resize(N, vector<T>(LOG + 1));
        for (int i = 0; i < N; ++i) jmp[i][0] = v[i];
        for (int j = 1; j <= LOG; ++j) {
            for (int i = 0; i + (1<<j) <= N; ++i) {
                jmp[i][j] = cmb(jmp[i][j-1], jmp[i+(1<<(j-1))][j-1]);
            }
        }
    }
    T query(int l, int r) {
        int d = __lg(r-l+1);
        return cmb(jmp[l][d], jmp[r-(1<<d)+1][d]);
    }
};
```

BIT.h

Description: Computes partial sums $a[0] + a[1] + \dots + a[\text{pos} - 1]$, and updates single elements $a[i]$, taking the difference between the old and new value. 0-Indexed.
Time: Both operations are $\mathcal{O}(\log N)$.

29f859, 26 lines

```
template<class T> struct BIT {
    int N; vector<T> bit;
    BIT(int _N) {
        N = _N;
        bit.resize(N);
    }
    void add(int p, T x) {
        for (++p; p <= N; p += p&-p) bit[p - 1] += x;
    }
    T sum(int l, int r) { return sum(r+1)-sum(l); }
    T sum(int r) {
        T res = 0;
        for (; r; r -= r&-r) res += bit[r - 1];
        return res;
    }
    int lower_bound(T sum) {
        if (sum <= 0) return -1;
        int pos = 0;
        for (int pw = 1 << 25; pw; pw >>= 1) {
            int npos = pos + pw;
            if (npos <= N && bit[npos - 1] < sum)
                pos = npos, sum -= bit[pos - 1];
        }
        return pos;
    }
};
```

SegmentTree.h

Description: 1D point update and range query where cmb is any associative operation. $\text{seg}[1] == \text{query}(0, N-1)$.
Time: $\mathcal{O}(\log N)$

a09d9c, 36 lines

template<class T> struct SegmentTree {
 const T ID{};
 T cmb(T a, T b) { return a + b; }
 int N = 1; vector<T> seg;
 SegmentTree(int _N) {
 while (N < _N) N *= 2;
 seg.assign(2*N, ID);
 }
 void pull(int p) { seg[p] = cmb(seg[2*p], seg[2*p+1]); }
 void update(int p, T val) { // set val at position p
 seg[p += N] = val;
 for (p /= 2; p > 0; p /= 2) pull(p);
 }
 T query(int l, int r) { // zero-indexed, inclusive
 T lf = ID, rf = ID;
 for (l += N, r += N + 1; l < r; l /= 2, r /= 2) {
 if (l & 1) lf = cmb(lf, seg[l++]);
 if (r & 1) rf = cmb(seg[--r], rf);
 }
 return cmb(lf, rf);
 }
 // int first_at_least(int lo, int val, int ind, int l, int r)
 // ↪ { // if seg stores max across range
 // if (r < lo || val > seg[ind]) return -1;
 // if (l == r) return l;
 // int m = (l+r)/2;
 // int res = first_at_least(lo, val, 2*ind, l, m); if (res !=
 // ↪-1) return res;
 // return first_at_least(lo, val, 2*ind+1, m+1, r);
 // }
 // int k_th_one(int k, int ind, int l, int r) {
 // if (l == r) return l;
 // int m = (l + r) / 2;
 // int lc = 2*ind, rc = 2*ind+1;
 // if (seg[lc] >= k) return k_th_one(k, lc, l, m);
 // else return k_th_one(k-seg[lc], rc, m+1, r);
 // }
};

LazySegmentTree.h

Description: 1D range increment and sum query.
Usage: LazySeg<int64_t, 1<<20> T; T.update(l, r, val);
Time: $\mathcal{O}(\log N)$

e0742f, 42 lines

```
template<class T, int SZ>struct LazySeg {
    // SZ must be power of 2
    static_assert(__builtin_popcount(SZ) == 1);
    const T ID{};
    T cmb(T a, T b) { return a + b; }
    T seg[2*SZ], lazy[2*SZ];
    LazySeg() {
        for (int i = 0; i < 2*SZ; ++i) seg[i] = lazy[i] = ID;
    }
    // modify values for current node
    void push(int ind, int L, int R) {
        seg[ind] += (R-L+1) * lazy[ind]; // dependent on operation
        if (L != R) for (int i = 0; i < 2; ++i) lazy[2*ind+i] +=
            ↪lazy[ind];
        lazy[ind] = 0;
    }
    void pull(int ind) {
        seg[ind] = cmb(seg[2*ind], seg[2*ind+1]);
    }
    void build() {
        for (int i = SZ-1; i >= 1; --i) pull(i);
    }
};
```

```
    }
    void update(int lo, int hi, T inc, int ind = 1, int L = 0,
        ↪int R = SZ-1) {
        push(ind, L, R);
        if (hi < L || R < lo) return;
        if (lo <= L && R <= hi) {
            lazy[ind] = inc;
            push(ind, L, R);
            return;
        }
        int M = (L+R) / 2;
        update(lo, hi, inc, 2*ind, L, M);
        update(lo, hi, inc, 2*ind+1, M+1, R);
        pull(ind);
    }
    T query(int lo, int hi, int ind = 1, int L = 0, int R = SZ-1)
        ↪ {
        push(ind, L, R);
        if (lo > R || L > hi) return ID;
        if (lo <= L && R <= hi) return seg[ind];
        int M = (L+R) / 2;
        return cmb(query(lo, hi, 2*ind, L, M), query(lo, hi, 2*ind
            ↪+1, M+1, R));
    }
};
```

3.3 2D Range Queries

PrefixSum2D.h

Description: calculates rectangle sums in constant time

65b070, 17 lines

```
template<typename T> struct PrefixSum2D {
    vector<vector<T>> sum;
    PrefixSum2D(const vector<vector<T>>& &v) {
        int n = v.size(), m = v[0].size();
        sum.assign(n+1, vector<T>(m+1, T{}));
        for (int i = 0; i < n; ++i) {
            for (int j = 0; j < m; ++j) {
                sum[i+1][j+1] = v[i][j]
                    - sum[i][j] + sum[i+1][j] + sum[i][j+1];
            }
        }
    }
    T query(int x1, int y1, int x2, int y2) {
        return sum[x2][y2] + sum[x1-1][y1-1]
            - sum[x1-1][y2] - sum[x2][y1-1];
    }
};
```

Number Theory (4)

4.1 Modular Arithmetic

$$a^{b^c} \bmod m = a^{b^c \bmod (m-1)} \bmod m$$

ModIntShort.h

Description: Modular arithmetic. Assumes MOD is prime.
Usage: mint a = MOD+5; inv(a); // 400000003

7c1e94, 30 lines

```
struct mint {
    int v;
    static const int MOD = 1E9 + 7;
    explicit operator int() const { return v; }
    mint() : v(0) {}
    mint(int64_t _v) : v((int)(_v % MOD)) { v += (v < 0) * MOD; }
    mint &operator+=(mint o) {
        if ((v += o.v) >= MOD) v -= MOD;
    }
};
```

```
        return *this;
    }
    mint &operator--=(mint o) {
        if ((v -= o.v) < 0) v += MOD;
        return *this;
    }
    mint &operator*=(mint o) {
        v = int((int64_t)v * o.v % MOD);
        return *this;
    }
    friend mint pow(mint a, int64_t p) {
        assert(p >= 0);
        return p==0? 1:pow(a*a, p/2)*(p&1? a:1);
    }
    friend mint inv(mint a) {
        assert(a.v != 0);
        return pow(a, MOD - 2);
    }
    friend mint operator+(mint a, mint b) { return a += b; }
    friend mint operator-(mint a, mint b) { return a -= b; }
    friend mint operator*(mint a, mint b) { return a *= b; }
};
```

ModFactInt.h

Description: Combinations modulo a prime MOD . Assumes $2 \leq N \leq MOD$.

Usage: F.gen(100), F.C(6, 4); // 15

Time: $\mathcal{O}(N)$

```
constexpr int MOD = 1E9 + 7;
struct {
    vector<int> inv, fac, ifac;
    void gen(int N) {
        inv.resize(N); fac.resize(N); ifac.resize(N);
        inv[1] = fac[0] = ifac[0] = 1;
        for (int i = 2; i < N; i++){
            inv[i] = int(MOD-(int64_t)MOD/i*inv[MOD%i]%MOD);
        }
        for (int i = 1; i < N; i++) {
            fac[i] = int(1LL*fac[i-1]*i%MOD);
            ifac[i] = int(1LL*ifac[i-1]*inv[i]%MOD);
        }
    }
    int P(int n, int r) { // nPr
        if (n < r || r < 0) return 0;
        return 1LL*fac[n]*ifac[n-r]%MOD;
    }
    int C(int n, int r) { // nCr
        if (n < r || r < 0) return 0;
        return 1LL*fac[n]*ifac[r]%MOD*ifac[n-r]%MOD;
    }
} F;
```

ModMulLL.h

Description: Multiply two 64-bit integers mod another if 128-bit is not available. modMul is equivalent to (ul) (__int128(a)*b%mod). Works for $0 \leq a, b < mod < 2^{63}$. NOTE : Only works if system supports 80-bit floating point.

```
using ul = uint64_t;
ul modMul(ul a, ul b, const ul mod){
    return __int128(a) * __int128(b) % mod;
}
ul modPow(ul a, ul b, const ul mod) {
    if (b == 0) return 1;
    ul res = modPow(a, b/2, mod);
    res = modMul(res, res, mod);
    return b&1 ? modMul(res, a, mod) : res;
}
```

FastMod.h

Description: Barrett reduction computes $a\%b$ about 4 times faster than usual where $b > 1$ is constant but not known at compile time. Division by b is replaced by multiplication by m and shifting right 64 bits.

```
using ul = uint64_t;
struct FastMod {
    ul b, m;
    FastMod(ul b) : b(b), m(-1ULL / b) {}
    ul reduce(ul a) {
        ul q = (ul)((__uint128_t(m) * a) >> 64), r = a - q*b;
        return r - (r>=b) * b;
    }
};
```

ModSqrt.h

Description: Tonelli-Shanks algorithm for square roots mod a prime. -1 if doesn't exist.

Usage: sqrt(mi*((11)1e10)); // 100000

Time: $\mathcal{O}(\log^2(MOD))$

```
"ModInt.h"
using T = int;
T sqrt(mi a) {
    mi p = pow(a, (MOD-1)/2);
    if (p.v != 1) return p.v == 0 ? 0 : -1;
    T s = MOD-1; int r = 0; while (s%2 == 0) s /= 2, ++r;
    mi n = 2; while (pow(n, (MOD-1)/2).v == 1) n = T(n)+1;
    // n non-square, ord(g)=2^r, ord(b)=2^m, ord(g)=2^r, m<r
    for (mi x = pow(a, (s+1)/2), b = pow(a, s), g = pow(n, s);) {
        if (b.v == 1) return min(x.v, MOD-x.v); // x^2=ab
        int m = 0; for (mi t = b; t.v != 1; t *= t) ++m;
        rep(r-m-1) g *= g; // ord(g)=2^m+1
        x *= g, g *= g, b *= g, r = m; // ord(g)=2^m, ord(b)<2^m
    }
}
```

ModSum.h

Description: Counts # of lattice points (x, y) in the triangle $1 \leq x, 1 \leq y, ax + by \leq s \pmod{2^{64}}$ and related quantities.

Time: $\mathcal{O}(\log ab)$

```
using ul = uint64_t;
ul sum2(ul n) { return n/2*((n-1)|1); } // sum(0..n-1)
// \return |\{(x,y) | 1 <= x, 1 <= y, a*x+b*y <= S\}|
// = sum_{i=1}^S |\{qs\} (S-a*i)/b|
ul triSum(ul a, ul b, ul s) { assert(a > 0 && b > 0);
    ul qs = s/a, rs = s%a; // ans = sum_{i=0}^{qs-1} (i*a+rs)/b
    ul ad = a/b*sum2(qs)+rs/b*qs; a %= b, rs %= b;
    return ad+(a?triSum(b,a,a+qs+rs):0); // reduce if a >= b
} // then swap x and y axes and recurse

// \return sum_{x=0}^{n-1} (a*x+b)/m
// = |\{(x,y) | 0 < m*y <= a*x+b < a*n+b\}|
// assuming a*n+b does not overflow
ul divSum(ul n, ul a, ul b, ul m) { assert(m > 0);
    ul extra = b/m*n; b %= m;
    return extra+(a?triSum(m,a,a+n*b):0); }
// \return sum_{x=0}^{n-1} (a*x+b)%m
ul modSum(ul n, ll a, ll b, ul m) { assert(m > 0);
    a = (a%m+m)%m, b = (b%m+m)%m;
    return a*sum2(n)+b*n-m*divSum(n,a,b,m); }
```

4.2 Primality

4.2.1 Primes

$p = 962592769$ is such that $2^{21} \mid p - 1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a , except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

4.2.2 Divisors

$$\sum_{d|n} d = O(n \log \log n).$$

The number of divisors of n is at most around 100 for $n < 5e4$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

Let $n = \prod_i p_i^{e_i}$ be the prime factorization of n .

Number of Divisors: $d(n) = \prod_i (e_i + 1)$.

$$\text{Sum of Divisors: } \sigma(n) = \prod_i \frac{p_i^{e_i+1} - 1}{p_i - 1}.$$

Product of Divisors:

Case 1: n is a perfect square. Then $\prod_{d|n} d = \sqrt{n}^{d(n)}$.

Case 2: n is not a perfect square. Then $\prod_{d|n} d = n^{d(n)/2}$.

Euler's Totient Function:
 $\phi(n) = n \prod_i (1 - \frac{1}{p_i}) = \prod_i (p_i^{a_i} - p_i^{a_i-1})$.

Dirichlet Convolution: Given a function $f(x)$, let

$$(f * g)(x) = \sum_{d|x} g(d)f(x/d).$$

If the partial sums $s_{f*g}(n), s_g(n)$ can be computed in $O(1)$ and $s_f(1 \dots n^{2/3})$ can be computed in $O\left(n^{2/3}\right)$ then all $s_f\left(\frac{n}{d}\right)$ can as well. Use

$$s_{f*g}(n) = \sum_{d=1}^n g(d)s_f(n/d).$$

If $f(x) = \mu(x)$ then $g(x) = 1, (f * g)(x) = (x == 1)$, and $s_f(n) = 1 - \sum_{i=2}^n s_f(n/i)$.

If $f(x) = \phi(x)$ then $g(x) = 1, (f * g)(x) = x$, and $s_f(n) = \frac{n(n+1)}{2} - \sum_{i=2}^n s_f(n/i)$.

Let $s(x) = \sum_{i=1}^x \phi(i)$. Then

$$s(n) = \frac{n(n+1)}{2} - \sum_{i=2}^n s\left(\left\lfloor \frac{n}{i} \right\rfloor\right)$$

can be computed in faster than $\Theta(n)$.

FactorBasic.h

Description: Factors integers.

Time: $\mathcal{O}\left(\sqrt{N}\right)$ 8b0341, 50 lines

```
inline namespace factorBasic {
template<class T> vector<pair<T, int>>> factor(T x) {
    vector<pair<T, int>>> primes;
    for (T i = 2; i*i <= x; ++i) if (x%i == 0) {
        int t = 0;
        while (x%i == 0) x /= i, t++;
        primes.push_back({i, t});
    }
    if (x > 1) primes.push_back({x, 1});
    return primes;
}

/* Note:
 * number of operations needed s.t.
 *      phi(phi(...phi(n)...))=1
 * is O(log n).
 * Euler's theorem: a^{phi(p)}\equiv 1 (mod p), gcd(a,p)=1
 */
template<class T> T phi(T x) {
    for (auto& [p, exp]: factor(x)) x -= x/p;
    return x;
}

template<class T> void tour(vector<pair<T, int>>> &v, vector<T>
    &V, int ind, T cur) {
    if (ind == (int)size(v)) V.push_back(cur);
    else {
        T mul = 1;
        for (int i = 0; i <= v[ind].second; i++) {
            tour(v, V, ind+1, cur*mul);
            mul *= v[ind].first;
        }
    }
}

template<class T> vector<T> getDiv(T x) {
    auto v = factor(x);
    vector<T> V;
    tour(v, V, 0, (T)1);
    ranges::sort(V); return V;
}

template<class T> vector<T> getDiv(T x) {
    vector<T> V;
    for (T i = 1; i*i <= x; ++i) if (x%i == 0) {
        V.push_back(i);
        if (x/i != i) V.push_back(x/i);
    }
    ranges::sort(V); return V;
}
```

Sieve.h

Description: Tests primality up to N . Runs faster if only odd indices are stored.

Time: $\mathcal{O}\left(N\log\log N\right)$ or $\mathcal{O}\left(N\right)$ fa9967, 27 lines

```
template<int N> struct Sieve {
    bitset<N> is_prime; vector<int> primes;
    Sieve() {
        is_prime.set(); is_prime[0] = is_prime[1] = 0;
        for (int i = 4; i < N; i += 2) is_prime[i] = 0;
        for (int i = 3; i*i < N; i += 2) if (is_prime[i])
            for (int j = i*i; j < N; j += i*2) is_prime[j] = 0;
    }
};
```

```
for (int i = 0; i < N; i++) if (is_prime[i]) primes.
    <-push_back(i);
}
int spf[N];
Sieve() {
    for (int i = 1; i < N; i += 2) spf[i] = i;
    for (int i = 2; i < N; i += 2) spf[i] = 2;
    for (int i = 3; i*i < N; i += 2) if (spf[i] == i)
        for (int j = i*i; j < N; j += i*2) if (spf[j] == j) spf[j]
            <-= i;
}
// int spf[N];
// Sieve() { // above is faster
//     for (int i = 2; i < N; i++) {
//         if (spf[i] == 0) spf[i] = i, primes.push_back(i);
//         for (int p: primes) {
//             if (p > spf[i] || i*p >= N) break;
//             spf[i*p] = p;
//         }
//     }
// }
```

ETF.h

Description: Computes Euler Totient function

Time: $\mathcal{O}\left(N\log\log N\right)$ 7549e9, 11 lines

```
template<int N> struct ETF {
    int phi[N];
    ETF() {
        for (int i = 0; i < N; i++) phi[i] = i;
        for (int i = 2; i < N; i++) if (phi[i] == i) {
            for (int j = i; j < N; j += i) {
                phi[j] -= phi[j] / i;
            }
        }
    }
};
```

MultiplicativePrefixSums.h

Description: $\sum_{i=1}^N f(i)$ where $f(i) = \prod \text{val}[e]$ for each p^e in the factorization of i . Must satisfy $\text{val}[1] = 1$. Generalizes to any multiplicative function with $f(p) = p^{\text{fixed power}}$.

Time: $\mathcal{O}\left(\sqrt{N}\right)$

```
"Sieve.h"
vmi val;
mi get_prefix(ll N, int p = 0) {
    mi ans = N;
    for (; S.primes.at(p) <= N / S.primes.at(p); ++p) {
        ll new_N = N / S.primes.at(p) / S.primes.at(p);
        for (int idx = 2; new_N; ++idx, new_N /= S.primes.at(p)) {
            ans += (val.at(idx) - val.at(idx - 1))
                * get_prefix(new_N, p + 1);
        }
    }
    return ans;
}
```

PrimeCnt.h

Description: Counts number of primes up to N . Can also count sum of primes.

Time: $\mathcal{O}\left(N^{3/4}/\log N\right)$, 60ms for $N = 10^{11}$, 2.5s for $N = 10^{13}$ c04e96, 20 lines

```
ll count_primes(ll N) { // count_primes(1e13) == 346065536839
    if (N <= 1) return 0;
    int sq = (int)sqrt(N);
    vl big_ans((sq+1)/2), small_ans(sq+1);
```

```
FOR(i,1,sq+1) small_ans[i] = (i-1)/2;
FOR(i,sz(big_ans)) big_ans[i] = (N/(2*i+1)-1)/2;
vb skip(sq+1); int prime_cnt = 0;
for (int p = 3; p <= sq; p += 2) if (!skip[p]) { // primes
    for (int j = p; j <= sq; j += 2*p) skip[j] = 1;
    FOR(j,min((ll)sz(big_ans), (N/p/p+1)/2)) {
        ll prod = (ll)(2*j+1)*p;
        big_ans[j] -= (prod > sq ? small_ans[(double)N/prod]
            : big_ans[prod/2])-prime_cnt;
    }
    for (int j = sq, q = sq/p; q >= p; --q) for (; j >= q*p; --j)
        small_ans[j] -= small_ans[q]-prime_cnt;
    ++prime_cnt;
}
return big_ans[0]+1;
}
```

MillerRabin.h

Description: Deterministic primality test, works up to 2^{64} . For larger numbers, extend A randomly.

"ModMulLL.h" 329b76, 13 lines

```
bool prime(ul n) { // not ll!
    if (n < 2 || n % 6 % 4 != 1) return n - 2 < 2;
    ul s = __builtin_ctzll(n - 1), d = n >> s;
    ul A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
    for (auto a : A) {
        ul p = modPow(a, d, n), i = s;
        while (p != 1 && p != n-1 && a % n && i--) {
            p = modMul(p, p, n);
        }
        if (p != n-1 && i != s) return false;
    }
    return true;
}
```

FactorFast.h

Description: Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).

Time: $\mathcal{O}\left(N^{1/4}\right)$, less for numbers with small factors

"MillerRabin.h", "ModMulLL.h" 99cf33, 16 lines

```
ul pollard(ul n) { // return some nontrivial factor of n
    auto f = [n](ul x) { return modMul(x, x, n) + 1; };
    ul x = 0, y = 0, t = 30, prd = 2, i = 1, q;
    while (t++ % 40 || gcd(prd, n) == 1) {
        if (x == y) x = ++i, y = f(x);
        if ((q = modMul(prd, max(x, y)-min(x, y), n)) prd = q;
            x = f(x), y = f(f(y)));
    }
    return gcd(prd, n);
}
void factor_rec(ul n, map<ul, int>& cnt) {
    if (n == 1) return;
    if (prime(n)) { ++cnt[n]; return; }
    ul u = pollard(n);
    factor_rec(u, cnt), factor_rec(n/u, cnt);
}
```

4.3 Euclidean Algorithm

4.3.1 Bézout's identity

For $a \neq, b \neq 0$, then $d = \gcd(a, b)$ is the smallest positive integer for which there are integer solutions to

$ax + by = d$

If (x, y) is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right), \quad k \in \mathbb{Z}$$

FracInterval.h

Description: Given fractions $a < b$ with non-negative numerators and denominators, finds fraction f with lowest denominator such that $a < f < b$. Should work with all numbers less than 2^{62} .

```
1860f3, 6 lines
pl bet(pl a, pl b) {
  ll num = a.f/a.s; a.f -= num*a.s, b.f -= num*b.s;
  if (b.f > b.s) return {1+num,1};
  auto x = bet({b.s,b.f},{a.s,a.f});
  return {x.s+num*x.f,x.f};
}
```

Euclid.h

Description: Generalized Euclidean algorithm. euclid and invGeneral work for $A, B < 2^{62}$.

```
8bab06, 13 lines
// For A,B>=0, finds (x,y) s.t.
pair<int64_t, int64_t> euclid(int64_t A, int64_t B) {
  // Ax+By=gcd(A,B), |Ax|,|By|<=AB/gcd(A,B)
  if (!B) return {1, 0};
  pair<int64_t, int64_t> p = euclid(B, A%B);
  return {p.second, p.first - A/B*p.second};
}
// find x in [0,B) such that Ax=1 mod B
int64_t invGeneral(int64_t A, int64_t B) {
  pair<int64_t, int64_t> p = euclid(A, B);
  assert(p.first*A + p.second*B == 1);
  return p.first + (p.first<0)*B;
} // must have gcd(A,B)=1
```

CRT.h

Description: Chinese Remainder Theorem. $a.f \pmod{a.s}, b.f \pmod{b.s} \implies ? \pmod{\text{lcm}(a.s, b.s)}$. Should work for $ab < 2^{62}$.

```
"Euclid.h" 23df64, 10 lines
pl CRT(pl a, pl b) { assert(0 <= a.f && a.f < a.s && 0 <= b.f
  <-&& b.f < b.s);
  if (a.s < b.s) swap(a,b); // will overflow if b.s^2 > 2^{62}
  ll x,y; tie(x,y) = euclid(a.s,b.s);
  ll g = a.s*x+b.s*y, l = a.s/g*b.s;
  if ((b.f-a.f)%g) return {-1,-1}; // no solution
  // ?*a.s+a.f \equiv b.f \pmod{b.s}
  // ?=(b.f-a.f)/g*(a.s/g)^{-1} \pmod{b.s/g}
  x = (b.f-a.f)%b.s*x%b.s/g*a.s+a.f;
  return {x+(x<0)*l,1};
}
```

ModArith.h

Description: Statistics on mod'ed arithmetic series. minBetween and minRemainder both assume that $0 \leq L \leq R < B, AB < 2^{62}$

```
f68a6d, 40 lines
ll minBetween(ll A, ll B, ll L, ll R) {
  // min x s.t. exists y s.t. L <= A*x-B*y <= R
  A %= B;
  if (L == 0) return 0;
  if (A == 0) return -1;
  ll k = cdiv(L,A); if (A*k <= R) return k;
  ll x = minBetween(B,A,A-R%A,A-L%A); // min x s.t. exists y
  // s.t. -R <= Bx-Ay <= -L
  return x == -1 ? x : cdiv(B*x+L,A); // solve for y
}

// find min((Ax+C)%B) for 0 <= x <= M
```

FracInterval Euclid CRT ModArith

```
// aka find minimum non-negative value of A*x-B*y+C
// where 0 <= x <= M, 0 <= y
ll minRemainder(ll A, ll B, ll C, ll M) {
  assert(A >= 0 && B > 0 && C >= 0 && M >= 0);
  A %= B, C %= B; ckmin(M,B-1);
  if (A == 0) return C;
  if (C >= A) { // make sure C<A
    ll ad = cdiv(B-C,A);
    M -= ad; if (M < 0) return C;
    C += ad*A-B;
  }
  ll q = B/A, new_B = B%A; // new_B < A
  if (new_B == 0) return C; // B=q*A

  // now minimize A*x-new_B*y+C
  // where 0 <= x,y and x+q*y <= M, 0 <= C < new_B < A
  // q*y -> C-new_B*y
  if (C/new_B > M/q) return C-M/q*new_B;
  M -= C/new_B*q; C %= new_B; // now C < new_B

  // given y, we can compute x = ceil[(B-q*A)*y-C]/A]
  // so x+q*y = ceil((B*y-C)/A) <= M
  ll max_Y = (M*A+C)/B; // must have y <= max_Y
  ll max_X = cdiv(new_B*max_Y-C,A); // must have x <= max_X
  if (max_X*A-new_B*max_Y+C >= new_B) --max_X;
  // now we can remove upper bound on y
  return minRemainder(A,new_B,C,max_X);
}
```

4.4 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2), \quad b = k \cdot (2mn), \quad c = k \cdot (m^2 + n^2),$$

with $m > n > 0, k > 0, m \perp n$, and either m or n even.

4.5 Lifting the Exponent

For $n > 0, p$ prime, and ints x, y s.t. $p \nmid x, y$ and $p \mid x - y$:

- $p \neq 2$ or $p = 2, 4 \mid x - y \implies v_p(x^n - y^n) = v_p(x - y) + v_p(n)$.
- $p = 2, 2 \mid n \implies v_2(x^n - y^n) = v_2((x^2)^{n/2} - (y^2)^{n/2})$.

Combinatorial (5)

5.1 Permutations

5.1.1 Cycles

Let $g_S(n)$ be the number of n -permutations whose cycle lengths all belong to the set S . Then

$$\sum_{n=0}^\infty g_S(n) \frac{x^n}{n!} = \exp \left(\sum_{n \in S} \frac{x^n}{n} \right)$$

5.1.2 Burnside's lemma

Given a group G of symmetries and a set X , the number of elements of X up to symmetry equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g \cdot x = x$).

If $f(n)$ counts “configurations” (of some sort) of length n , we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k|n} f(k) \phi(n/k).$$

5.2 Partitions and subsets

5.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1, \quad p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k - 1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	$\sim 2\text{e}5$	$\sim 2\text{e}8$

5.2.2 Lucas' Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + \dots + n_1 p + n_0$ and $m = m_k p^k + \dots + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^k \binom{n_i}{m_i} \pmod{p}$.

5.3 General purpose numbers

5.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able). $B[0, \dots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \dots]$

Sums of powers:

$$\sum_{i=1}^n i^m = \frac{1}{m+1} \sum_{k=0}^m \binom{m+1}{k} B_k (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\begin{aligned} \sum_{i=m}^\infty f(i) &= \int_m^\infty f(x) dx - \sum_{k=1}^\infty \frac{B_k}{k!} f^{(k-1)}(m) \\ &\approx \int_m^\infty f(x) dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m)) \end{aligned}$$

5.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n, k) = c(n-1, k-1) + (n-1)c(n-1, k), \quad c(0, 0) = 1$$
$$\sum_{k=0}^n c(n, k) x^k = x(x+1) \dots (x+n-1)$$

$$c(8, k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$$
$$c(n, 2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \dots$$

5.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j :s s.t. $\pi(j) > \pi(j + 1)$, $k + 1$ j :s s.t. $\pi(j) \geq j$, k j :s s.t. $\pi(j) > j$.

$$E(n,k) = (n - k)E(n - 1, k - 1) + (k + 1)E(n - 1, k)$$

$$E(n,0) = E(n,n - 1) = 1$$

$$E(n,k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n$$

5.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n,k) = S(n - 1, k - 1) + kS(n - 1, k)$$

$$S(n,1) = S(n,n) = 1$$

$$S(n,k) = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$$

5.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \dots$ For p prime,

$$B(p^m + n) \equiv mB(n) + B(n + 1) \pmod{p}$$

5.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i : $n_1 n_2 \cdots n_k n^{k-2}$
with degrees d_i : $(n - 2)! / ((d_1 - 1)! \cdots (d_n - 1)!)$

5.3.7 Catalan numbers

$$C_n = \frac{1}{n + 1} \binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n + 1} = \frac{(2n)!}{(n + 1)!n!}$$

$$C_0 = 1, C_{n+1} = \frac{2(2n + 1)}{n + 2} C_n, C_{n+1} = \sum C_i C_{n-i}$$

$$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \dots$$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n + 1$ leaves (0 or 2 children).
- ordered trees with $n + 1$ vertices.
- ways a convex polygon with $n + 2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

5.4 Young Tableaux

Let a **Young diagram** have shape $\lambda = (\lambda_1 \geq \cdots \geq \lambda_k)$, where λ_i equals the number of cells in the i -th (left-justified) row from the top. A **Young tableau** of shape λ is a filling of the $n = \sum \lambda_i$ cells with a permutation of $1 \dots n$ such that each row and column is increasing.

Hook-Length Formula: For the cell in position (i, j) , let $h_\lambda(i, j) = |\{(I, J) | i \leq I, j \leq J, (I = i \text{ or } J = j)\}|$. The number of Young tableaux of shape λ is equal to $f^\lambda = \frac{n!}{\prod h_\lambda(i, j)}$.

Schensted’s Algorithm: converts a permutation σ of length n into a pair of Young Tableaux $(S(\sigma), T(\sigma))$ of the same shape. When inserting $x = \sigma_i$,

1. Add x to the first row of S by inserting x in place of the largest y with $x < y$. If y doesn’t exist, push x to the end of the row, set the value of T at that position to be i , and stop.
2. Add y to the second row using the same rule, keep repeating as necessary.

All pairs $(S(\sigma), T(\sigma))$ of the same shape correspond to a unique σ , so $n! = \sum (f^\lambda)^2$. Also, $S(\sigma^R) = S(\sigma)^T$.

Let $d_k(\sigma), a_k(\sigma)$ be the lengths of the longest subseqs which are a union of k decreasing/ascending subseqs, respectively. Then $a_k(\sigma) = \sum_{i=1}^k \lambda_i, d_k(\sigma) = \sum_{i=1}^k \lambda_i^*$, where λ_i^* is size of the i -th column.

5.5 Other

```
DeBruijnSequence.h
Description: Given alphabet [0, k) constructs a cyclic string of length k^n
that contains every length n string as substring.
vector<int> deBruijnSequence(int k, int n) {
    if (k == 1) return {0};
    vector<int> seq, aux(n+1);
    auto gen = [&](this auto&& self, int t, int p)->void {
        if (t > n && n%p == 0) for (int i = 1; i <= p; i++) {
            seq.push_back(aux[i]);
        }
        else if (t <= n) {
            aux[t] = aux[t-p]; self(t+1, p);
            while (++aux[t] < k) self(t+1, t);
        }
    };
    gen(1, 1); return seq;
}
```

```
NimProduct.h
Description: Product of numbrs is associative, commutative, and dis-
tributive over addition (xor). Forms finite field of size 2^{2^k}. Defined by
ab = mex({a'b + ab' + a'b' : a' < a, b' < b}). Application: Given 1D
coin turning games G_1, G_2 G_1 x G_2 is the 2D coin turning game defined as
follows. If turning coins at x_1, x_2, ..., x_m is legal in G_1 and y_1, y_2, ..., y_n
is legal in G_2, then turning coins at all positions (x_i, y_j) is legal assuming
that the coin at (x_m, y_n) goes from heads to tails. Then the grundy function
g(x, y) of G_1 x G_2 is g_1(x) x g_2(y).
Time: 64^2 xors per multiplication, memorize to speed up.
```

```
using ul = uint64_t;
struct Precalc {
    ul tmp[64][64], y[8][8][256];
    unsigned char x[256][256];
    Precalc() { // small nim products, all < 256
        F0R(i,256) F0R(j,256) x[i][j] = mult<8>(i,j);
        F0R(i,8) F0R(j,i+1) F0R(k,256)
            y[i][j][k] = mult<64>(prod2(8*i,8*j),k);
    }
    ul prod2(int i, int j) { // nim prod of 2^i, 2^j
        ul& u = tmp[i][j]; if (u) return u;
        if (!(i&j)) return u = 1ULL<<(i|j);
        int a = (i&j)&-(i&j); // a=2^k, consider 2^{2^k}
        return u=prod2(i^a,j)^prod2((i^a|(a-1),(j^a)|(i&(a-1))));
        // 2^{2^k}*2^{2^k} = 2^{2^k+2^{2^k-1}}
    } // 2^{2^i}*2^{2^j} = 2^{2^i+2^j} if i<j
    template<int L> ul mult(ul a, ul b) {
        ul c = 0; F0R(i,L) if (a>>i&1)
            F0R(j,L) if (b>>j&1) c ^= prod2(i,j);
        return c;
    }
};
const Precalc P;

struct nb { // nimber
    ul x; nb() { x = 0; }
    nb(ul _x): x(_x) {}
    explicit operator ul() { return x; }
    nb operator+(nb y) { return nb(x^y.x); }
    nb operator*(nb y) { return nb(P.multFast(x,y.x)); }
    friend nb pow(nb b, ul p) {
        nb res = 1; for (;p;p/=2,b=b*b) if (p&1) res = res*b;
        return res; } // b^{2^{2^A}-1}=1 where 2^{2^A} > b
    friend nb inv(nb b) { return pow(b,-2); }
};

MatroidIsect.h
Description: Computes a set of maximum size which is independent in both
graphic and colorful matroids, aka a spanning forest where no two edges are
of the same color. In general, construct the exchange graph and find a short-
est path. Can apply similar concept to partition matroid.
Usage: MatroidIsect<Gmat,Cmat> M(sz(ed),Gmat(ed),Cmat(col))
Time: O(GI^{1.5}) calls to oracles, where G is size of ground set and I is size
of independent set.
"DSU.h" d0051c, 51 lines

struct Gmat { // graphic matroid
    int V = 0; vpi ed; DSU D;
    Gmat(vpi _ed):ed(_ed) {
        map<int,int> m; each(t,ed) m[t.f] = m[t.s] = 0;
        each(t,m) t.s = V++;
        each(t,ed) t.f = m[t.f], t.s = m[t.s];
    }
    void clear() { D.init(V); }
    void ins(int i) { assert(D.unite(ed[i].f,ed[i].s)); }
    bool indep(int i) { return !D.sameSet(ed[i].f,ed[i].s); }
};
struct Cmat { // colorful matroid
    int C = 0; vi col; V<bool> used;
```

```
Cmat(vi col):col(col) {each(t,col) ckmax(C,t+1); }
void clear() { used.assign(C,0); }
void ins(int i) { used[col[i]] = 1; }
bool indep(int i) { return !used[col[i]]; }
};
template<class M1, class M2> struct MatroidIsect {
    int n; V<bool> iset; M1 m1; M2 m2;
    bool augment() {
        vi pre(n+1,-1); queue<int> q({n});
        while (sz(q)) {
            int x = q.ft; q.pop();
            if (iset[x]) {
                m1.clear(); F0R(i,n) if (iset[i] && i != x) m1.ins(i);
                F0R(i,n) if (!iset[i] && pre[i] == -1 && m1.indep(i))
                    pre[i] = x, q.push(i);
            } else {
                auto backE = [&]() { // back edge
                    m2.clear();
                    F0R(c,2)F0R(i,n)if((x==i||iset[i])&&(pre[i]==-1)==c){
                        if (!m2.indep(i))return c?pre[i]=x,q.push(i),i:-1;
                        m2.ins(i); }
                    return n;
                };
                for (int y; (y = backE()) != -1;) if (y == n) {
                    for(; x != n; x = pre[x]) iset[x] = !iset[x];
                    return 1; }
            }
        }
        return 0;
    }
    MatroidIsect(int n, M1 m1, M2 m2):n(n), m1(m1), m2(m2) {
        iset.assign(n+1,0); iset[n] = 1;
        m1.clear(); m2.clear(); // greedily add to basis
        R0F(i,n) if (m1.indep(i) && m2.indep(i))
            iset[i] = 1, m1.ins(i), m2.ins(i);
        while (augment());
    }
};
```

Numerical (6)

6.1 Arithmetic

Int128.h

Description: Operation on 128bit Integer

7855c4, 104 lines

```
struct i128 {
    __int128 value;

    // Constructors
    i128() : value(0) {}
    i128(int x) : value(x) {}
    i128(long long x) : value(x) {}
    i128(__int128 x) : value(x) {}

    // GCD
    static i128 gcd(const i128 &a, const i128 &b) {
        __int128 x = a.value, y = b.value;
        while (y != 0) {
            __int128 temp = y;
            y = x % y;
            x = temp;
        }
        return i128(x);
    }

    i128 sqrt() const {
        if (value < 0) return i128(0); // Handle negative input
```

```
if (value <= 1) return i128(value); // Handle 0 and 1

    __int128 low = 1;
    __int128 high = value;
    __int128 result = 0;

    while (low <= high) {
        __int128 mid = low + (high - low) / 2;
        __int128 quotient = value / mid; // Avoids overflow

        if (mid <= quotient) {
            // mid^2 <= value (mid is candidate)
            result = mid;
            low = mid + 1;
        } else {
            // mid^2 > value (adjust upper bound)
            high = mid - 1;
        }
    }

    return i128(result);
}

// Convert to string for output
string to_string() const {
    if (value == 0) return "0";
    __int128 tmp = value;
    bool neg = tmp < 0;
    if (neg) tmp = -tmp;
    string res;
    while (tmp) {
        res += (char)('0' + tmp % 10);
        tmp /= 10;
    }
    if (neg) res += '-';
    reverse(res.begin(), res.end());
    return res;
}

// Convert from string for input
static i128 from_string(const string &s) {
    __int128 res = 0;
    bool neg = false;
    size_t i = 0;
    if (s[0] == '-') {
        neg = true;
        i = 1;
    }
    for (; i < s.size(); i++) {
        res = res * 10 + (s[i] - '0');
    }
    return neg ? i128(-res) : i128(res);
}

// Operators
i128 operator+(const i128 &b) const { return i128(value + b.
    ↪value); }
i128 operator-(const i128 &b) const { return i128(value - b.
    ↪value); }
i128 operator*(const i128 &b) const { return i128(value * b.
    ↪value); }
i128 operator/(const i128 &b) const { return i128(value / b.
    ↪value); }
i128 operator%(const i128 &b) const { return i128(value % b.
    ↪value); }

bool operator<(const i128 &b) const { return value < b.value;
    ↪ }
```

```
bool operator>(const i128 &b) const { return value > b.value;
    ↪ }
bool operator<=(const i128 &b) const { return value <= b.
    ↪value; }
bool operator>=(const i128 &b) const { return value >= b.
    ↪value; }
bool operator==(const i128 &b) const { return value == b.
    ↪value; }
bool operator!=(const i128 &b) const { return value != b.
    ↪value; }

// Input/Output
friend ostream &operator<<(ostream &os, const i128 &b) {
    return os << b.to_string();
}

// Fix the conversion issue for integers (long long, int, etc
    ↪.)
friend istream &operator>>(istream &is, i128 &b) {
    string s;
    is >> s;
    b = from_string(s);
    return is;
}
};

6.2 Matrix
Matrix.h
Description: 2D matrix operations.
45a081, 57 lines
ModInt.h
using T = mi;
using Mat = vector<vector<T>>; // use array instead if tight TL

Mat makeMat(int r, int c) {
    return Mat(r, vector<T>(c));
}
Mat makeId(int n) {
    Mat m = makeMat(n, n);
    for (int i = 0; i < n; i++) m[i][i] = 1;
    return m;
}

Mat operator+(Mat& a, const Mat& b) {
    assert(sz(a) == sz(b) && sz(a[0]) == sz(b[0]));
    int x = size(a), y = size(a[0]);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < y; j++) {
            a[i][j] += b[i][j];
        }
    }
    return a;
}

Mat operator-(Mat& a, const Mat& b) {
    assert(sz(a) == sz(b) && sz(a[0]) == sz(b[0]));
    int x = size(a), y = size(a[0]);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < y; j++) {
            a[i][j] -= b[i][j];
        }
    }
    return a;
}

Mat operator*(const Mat& a, const Mat& b) {
    int x = size(a), y = size(a[0]), z = size(b[0]);
    assert(y == sz(b)); Mat c = makeMat(x, z);
    for (int i = 0; i < x; i++) {
```

```
    for (int j = 0; j < y; j++) {
        for (int k = 0; k < z; k++) {
            c[i][k] += a[i][j]*b[j][k];
        }
    }
}
return c;
}

Mat& operator+=(Mat& a, const Mat& b) { return a = a+b; }
Mat& operator+=(Mat& a, const Mat& b) { return a = a+b; }
Mat& operator-=(Mat& a, const Mat& b) { return a = a-b; }

Mat pow(Mat m, int64_t p) {
    int n = sz(m); assert(n == sz(m[0]) && p >= 0);
    Mat res = makeId(n);
    for (; p; p /= 2, m *= m) if (p&1) res *= m;
    return res;
}
```

MatrixInv.h
Description: Uses gaussian elimination to convert into reduced row echelon form and calculates determinant. For determinant via arbitrary modulus, use a modified form of the Euclidean algorithm because modular inverse may not exist. If you have computed $A^{-1} \pmod{p^k}$, then the inverse $\pmod{p^{2k}}$ is $A^{-1}(2I - AA^{-1})$.
Time: $\mathcal{O}(N^3)$, determinant of 1000×1000 matrix of modints in 1 second if you reduce # of operations by half

```
const db EPS = 1e-9; // adjust?
int getRow(V<V<db>>& m, int R, int i, int nex) {
    pair<db,int> bes{0,-1}; // find row with max abs value
    FOR(j,nex,R) ckmax(bes,{abs(m[j][i]),j});
    return bes.f < EPS ? -1 : bes.s; }
int getRow(V<vmi>& m, int R, int i, int nex) {
    FOR(j,nex,R) if (m[j][i] != 0) return j;
    return -1; }
pair<T,int> gauss(Mat& m) { // convert to reduced row echelon
    ↪form
    if (!sz(m)) return {1,0};
    int R = sz(m), C = sz(m[0]), rank = 0, nex = 0;
    T prod = 1; // determinant
    F0R(i,C) {
        int row = getRow(m,R,i,nex);
        if (row == -1) { prod = 0; continue; }
        if (row != nex) prod *= -1, swap(m[row],m[nex]);
        prod *= m[nex][i]; rank++;
        T x = 1/m[nex][i]; FOR(k,i,C) m[nex][k] *= x;
        F0R(j,R) if (j != nex) {
            T v = m[j][i]; if (v == 0) continue;
            FOR(k,i,C) m[j][k] -= v*m[nex][k];
        }
        nex++;
    }
    return {prod,rank};
}
Mat inv(Mat m) {
    int R = sz(m); assert(R == sz(m[0]));
    Mat x = makeMat(R,2*R);
    F0R(i,R) {
        x[i][i+R] = 1;
        F0R(j,R) x[i][j] = m[i][j];
    }
    if (gauss(x).s != R) return Mat();
    Mat res = makeMat(R,R);
    F0R(i,R) F0R(j,R) res[i][j] = x[i][j+R];
    return res;
}
```

MatrixTree.h
Description: Kirchhoff's Matrix Tree Theorem. Given adjacency matrix, calculates # of spanning trees.

```
"MatrixInv.h" 48363d, 11 lines
T numSpan(const Mat& m) {
    int n = sz(m); Mat res = makeMat(n-1,n-1);
    F0R(i,n) FOR(j,i+1,n) {
        mi ed = m[i][j]; res[i][i] += ed;
        if (j != n-1) {
            res[j][j] += ed;
            res[i][j] -= ed, res[j][i] -= ed;
        }
    }
    return gauss(res).f;
}
```

ShermanMorrison.h
Description: Calculates $(A + uv^T)^{-1}$ given $B = A^{-1}$. Not invertible if sum=0.

```
"MatrixInv.h" 3a3f34, 7 lines
void ad(Mat& B, const V<T>& u, const V<T>& v) {
    int n = sz(A); V<T> x(n), y(n);
    F0R(i,n) F0R(j,n)
        x[i] += B[i][j]*u[j], y[j] += v[i]*B[i][j];
    T sum = 1; F0R(i,n) F0R(j,n) sum += v[i]*B[i][j]*u[j];
    F0R(i,n) F0R(j,n) B[i][j] -= x[i]*y[j]/sum;
}
```

6.3 Polynomials

Poly.h
Description: Basic poly ops including division. Can replace T with double, complex.

```
"ModInt.h" cd218a, 73 lines
using T = mi; using poly = V<T>;
void remz(poly& p) { while (sz(p)&&p.bk==T(0)) p.pop_back(); }
poly REMZ(poly p) { remz(p); return p; }
poly rev(poly p) { reverse(all(p)); return p; }
poly shift(poly p, int x) {
    if (x >= 0) p.insert(begin(p),x,0);
    else assert(sz(p)+x >= 0), p.erase(begin(p),begin(p)-x);
    return p;
}
poly RSZ(const poly& p, int x) {
    if (x <= sz(p)) return poly(begin(p),begin(p)+x);
    poly q = p; q.rsz(x); return q; }
T eval(const poly& p, T x) { // evaluate at point x
    T res = 0; R0F(i,sz(p)) res = x*res+p[i];
    return res; }
poly dif(const poly& p) { // differentiate
    poly res; FOR(i,1,sz(p)) res.pb(T(i)*p[i]);
    return res; }
poly integ(const poly& p) { // integrate
    static poly invs{0,1};
    for (int i = sz(invs); i <= sz(p); ++i)
        invs.pb(-MOD/i*invs[MOD%i]);
    poly res(sz(p)+1); F0R(i,sz(p)) res[i+1] = p[i]*invs[i+1];
    return res;
}
```

```
poly& operator+=(poly& l, const poly& r) {
    l.rsz(max(sz(l),sz(r))); F0R(i,sz(r)) l[i] += r[i];
    return l; }
poly& operator-=(poly& l, const poly& r) {
    l.rsz(max(sz(l),sz(r))); F0R(i,sz(r)) l[i] -= r[i];
    return l; }
poly& operator*=(poly& l, const T& r) { each(t,l) t *= r;
    return l; }
```

```
poly& operator/=(poly& l, const T& r) { each(t,l) t /= r;
    return l; }
poly operator+(poly l, const poly& r) { return l += r; }
poly operator-(poly l, const poly& r) { return l -= r; }
poly operator-(poly l) { each(t,l) t *= -1; return l; }
poly operator*(poly l, const T& r) { return l *= r; }
poly operator*(const T& r, const poly& l) { return l*r; }
poly operator/(poly l, const T& r) { return l /= r; }
poly operator*(const poly& l, const poly& r) {
    if (!min(sz(l),sz(r))) return {};
    poly x(sz(l)+sz(r)-1);
    F0R(i,sz(l)) F0R(j,sz(r)) x[i+j] += l[i]*r[j];
    return x;
}
poly& operator*=(poly& l, const poly& r) { return l = l*r; }
```

```
pair<poly,poly> quoRemSlow(poly a, poly b) {
    remz(a); remz(b); assert(sz(b));
    T lst = b.bk, B = T(1)/lst; each(t,a) t *= B;
    each(t,b) t *= B;
    poly q(max(sz(a)-sz(b)+1,0));
    for (int dif; (dif=sz(a)-sz(b)) >= 0; remz(a)) {
        q[dif] = a.bk; F0R(i,sz(b)) a[i+dif] -= q[dif]*b[i]; }
    each(t,a) t *= lst;
    return {q,a}; // quotient, remainder
}
poly operator%(const poly& a, const poly& b) {
    return quoRemSlow(a,b).s; }
T resultant(poly a, poly b) { // R(A,B)
    // =b_m^n*prod_{j=1}^m A(mu_j)
    // =b_m^na_n^m*prod_{i=1}^nprod_{j=1}^m (mu_j-lambda_i)
    // =(-1)^{nm}a_n^m*prod_{i=1}^nB(lambda_i)
    // =(-1)^{nm}R(B,A)
    // Also, R(A,B)=b_m^{deg(A)-deg(A-CB)}R(A-CB,B)
    int ad = sz(a)-1, bd = sz(b)-1;
    if (bd <= 0) return bd < 0 ? 0 : pow(b.bk,ad);
    int pw = ad; a = a*b; pw -= (ad = sz(a)-1);
    return resultant(b,a)*pow(b.bk,pw)*T((bd&ad&1)?-1:1);
}
```

PolyInterpolate.h
Description: n points determine unique polynomial of degree $\leq n-1$. For numerical precision pick $v[k].f = c * \cos(k/(n-1) * \pi)$, $k = 0 \dots n-1$.
Time: $\mathcal{O}(n^2)$

```
"Poly.h" aada3a, 8 lines
poly interpolate(V<pair<T,T>> v) {
    poly res, tmp{1};
    F0R(i,sz(v)) { T prod = 1; // add one point at a time
        F0R(j,i) v[i].s -= prod*v[j].s, prod *= v[i].f-v[j].f;
        v[i].s /= prod; res += v[i].s*tmp; tmp *= poly{-v[i].f,1};
    } // add multiple of (x-v[0].f)*(x-v[1].f)*...*(x-v[i-1].f)
    return res;
}
```

FFT.h
Description: Multiply polynomials of ints for any modulus $< 2^{31}$. For XOR convolution ignore m within fft.
Time: $\mathcal{O}(N \log N)$. For $N = 10^6$, conv ~0.13ms, conv_general ~320ms.

```
"ModInt.h" 19ab20, 39 lines
// const int MOD = 998244353;
tcT> void fft(V<T>& A, bool invert = 0) { // NTT
    int n = sz(A); assert((T::mod-1)%n == 0); V<T> B(n);
    for(int b = n/2; b; b /= 2, swap(A,B)) { // w = n/b^th root
        T w = pow(T::rt(),(T::mod-1)/n*b), m = 1;
        for(int i = 0; i < n; i += b*2, m *= w) F0R(j,b) {
            T u = A[i+j], v = A[i+j+b]*m;
            B[i/2+j] = u+v; B[i/2+j+n/2] = u-v;
        }
    }
}
```

```
    }
    if (invert) { reverse(1+all(A));
        T z = inv(T(n)); each(t,A) t *= z; }
} // for NTT-able moduli
tcT> V<T> conv(V<T> A, V<T> B) {
    if (!min(sz(A),sz(B))) return {};
    int s = sz(A)+sz(B)-1, n = 1; for (; n < s; n *= 2);
    A.rsz(n), fft(A); B.rsz(n), fft(B);
    F0R(i,n) A[i] *= B[i];
    fft(A,1); A.rsz(s); return A;
}
template<class M, class T> V<M> mulMod(const V<T>& x, const V<T
    ↪>& y) {
    auto con = [] (const V<T>& v) {
        V<M> w(sz(v)); F0R(i,sz(v)) w[i] = (int)v[i];
        return w; };
    return conv(con(x), con(y));
} // arbitrary moduli
tcT> V<T> conv_general(const V<T>& A, const V<T>& B) {
    using m0 = mint<(119<<23)+1,62>; auto c0 = mulMod<m0>(A,B);
    using m1 = mint<(5<<25)+1, 62>; auto c1 = mulMod<m1>(A,B);
    using m2 = mint<(7<<26)+1, 62>; auto c2 = mulMod<m2>(A,B);
    int n = sz(c0); V<T> res(n); m1 r01 = inv(m1(m0::mod));
    m2 r02 = inv(m2(m0::mod)), r12 = inv(m2(m1::mod));
    F0R(i,n) { // a=remainder mod m0::mod, b fixes it mod m1::mod
        int a = c0[i].v, b = ((cl[i]-a)*r01).v,
            c = (((c2[i]-a)*r02-b)*r12).v;
        res[i] = (T(c)*m1::mod+b)*m0::mod+a; // c fixes m2::mod
    }
    return res;
}
```

PolyInvSimpler.h

Description: computes A^{-1} such that $AA^{-1} \equiv 1 \pmod{x^n}$. Newton's method: If you want $F(x) = 0$ and $F(Q_k) \equiv 0 \pmod{x^a}$ then $Q_{k+1} = Q_k - \frac{F(Q_k)}{F'(Q_k)} \pmod{x^{2a}}$ satisfies $F(Q_{k+1}) \equiv 0 \pmod{x^{2a}}$. Application: if $f(n),g(n)$ are the #s of forests and trees on n nodes then $\sum_{n=0}^\infty f(n)x^n = \exp\left(\sum_{n=1}^\infty \frac{g(n)}{n!}\right)$.
Usage: vmi v{1,5,2,3,4}; ps(exp(2*log(v,9),9)); // squares v
Time: $\mathcal{O}(N \log N)$. For $N = 5 \cdot 10^5$, inv~270ms, log ~350ms, exp~550ms
"FFT.h", "Poly.h" 6e5362, 30 lines

```
poly inv(poly A, int n) { // Q-(1/Q-A)/(-Q^{-2})
    poly B(inv(A[0]));
    for (int x = 2; x/2 < n; x *= 2)
        B = 2*B-RSZ(conv(RSZ(A,x),conv(B,B)),x);
    return RSZ(B,n);
}
poly sqrt(const poly& A, int n) { // Q-(Q^2-A)/(2Q)
    assert(A[0].v == 1); poly B{1};
    for (int x = 2; x/2 < n; x *= 2)
        B = inv(T(2))*RSZ(B+conv(RSZ(A,x),inv(B,x)),x);
    return RSZ(B,n);
}
// return {quotient, remainder}
pair<poly,poly> quoRem(const poly& f, const poly& g) {
    if (sz(f) < sz(g)) return {{},f};
    poly q = conv(inv(rev(g),sz(f)-sz(g)+1),rev(f));
    q = rev(RSZ(q,sz(f)-sz(g)+1));
    poly r = RSZ(f-conv(q,g),sz(g)-1); return {q,r};
}
poly log(poly A, int n) { assert(A[0].v == 1); // (ln A)' = A'/
    ↪A
    A.rsz(n); return integ(RSZ(conv(dif(A),inv(A,n-1)),n-1)); }
poly exp(poly A, int n) { assert(A[0].v == 0);
    poly B{1}, IB{1}; // inverse of B
    for (int x = 1; x < n; x *= 2) {
        IB = 2*IB-RSZ(conv(B,conv(IB,IB)),x);
```

```
    poly Q = dif(RSZ(A,x)); Q += RSZ(conv(IB,dif(B)-conv(B,Q))
        ↪,2*x-1);
    B = B+RSZ(conv(B,RSZ(A,2*x)-integ(Q)),2*x);
    }
    return RSZ(B,n);
}
```

6.4 Search

fstTrue.h

Description: Finds the first value x in [lo, hi] such that predicate f(x) is true.
Usage: int ans = fstTrue()
Time: $\mathcal{O}(\log(hi-lo))$ 2654fa, 9 lines

```
template<typename T, typename U> T fstTrue(T lo, T hi, U f) {
    ++hi;
    assert(lo <= hi);
    while (lo < hi) {
        T mid = lo + (hi - lo) / 2;
        f(mid) ? hi = mid : lo = mid + 1;
    }
    return lo;
};
```

lstTrue.h

Description: Finds the last value x in [lo, hi] such that predicate f(x) is true.
Usage: int ans = fstTrue()
Time: $\mathcal{O}(\log(hi-lo))$ 8aff85, 9 lines

```
template<typename T, typename U> T lstTrue(T lo, T hi, U f) {
    --lo;
    assert(lo <= hi);
    while (lo < hi) {
        T mid = lo + (hi - lo + 1) / 2;
        f(mid) ? lo = mid : hi = mid - 1;
    }
    return lo;
};
```

realTrue.h

Description: Binary search on continuous (floating-point) domain.
Usage: int ans = realTrue()
Time: $\mathcal{O}(\log(precision))$ — write 100 iterations and have a relax 68fcd, 7 lines

```
template<typename T, typename U>T realTrue(T lo, T hi, U f) {
    for (int i = 0; i < 100; ++i) {
        T mid = 0.5 * (hi + lo);
        f(mid) ? lo = mid : hi = mid;
    }
    return lo;
};
```

Ternary.h

Description: Ternary search to find the min of a unimodal function
Time: $\mathcal{O}(\log(N/eps))$ ef144f, 9 lines

```
template<typename U>
double Ternary(double lo, double hi, U f) {
    while (hi - lo > eps) {
        double l = lo + (hi - lo) / 3;
        double r = hi - (hi - lo) / 3;
        f(l) > f(r) ? lo = l : hi = r;
    }
    return lo;
}
```

6.5 Misc

LinearRecurrence.h

Description: Berlekamp-Massey. Computes linear recurrence C of order N for sequence s of $2N$ terms. $C[0] = 1$ and for all $i \geq sz(C) - 1$, $\sum_{j=0}^{sz(C)-1} C[j]s[i-j] = 0$.
Usage: LinRec L; L.init({0,1,1,2,3}); L.eval(5); L.eval(6); // 5, 8
Time: init $\Rightarrow \mathcal{O}(N|C|)$, eval $\Rightarrow \mathcal{O}(|C|^2 \log p)$ or faster with FFT
"Poly.h" 39ea71, 29 lines

```
struct LinRec {
    poly s, C, rC;
    void BM() {
        int x = 0; T b = 1;
        poly B; B = C = {1}; // B is fail vector
        F0R(i,sz(s)) { // update C after adding a term of s
            ++x; int L = sz(C), M = i+3-L;
            T d = 0; F0R(j,L) d += C[j]*s[i-j]; // [D^i]C*s
            if (d.v == 0) continue; // [D^i]C*s=0
            poly _C = C; T coef = d*inv(b);
            C.rsz(max(L,M)); F0R(j,sz(B)) C[j+x] -= coef*B[j];
            if (L < M) B = _C, b = d, x = 0;
        }
    }
    void init(const poly& _s) {
        s = _s; BM();
        rC = C; reverse(all(rC));
        C.erase(begin(C)); each(t,C) t *= -1;
    } // now s[i]=sum_{j=0}^{sz(C)-1}C[j]*s[i-j-1]
    poly getPow(ll p) { // get x^p mod rC
        if (p == 0) return {1};
        poly r = getPow(p/2); r = (r*r)%rC;
        return p&1?(r*poly{0,1})%rC:r;
    }
    T dot(poly v) { // dot product with s
        T ans = 0; F0R(i,sz(v)) ans += v[i]*s[i];
        return ans; } // get p-th term of rec
    T eval(ll p) { assert(p >= 0); return dot(getPow(p)); }
};
```

Integrate.h

Description: Integration of a function over an interval using Simpson's rule, exact for polynomials of degree up to 3. The error should be proportional to dif^4 , although in practice you will want to verify that the result is stable to desired precision when epsilon changes.
Usage: quad({}(db x) { return x*x+3*x+1; }, 2, 3) // 14.833 3ebaab, 6 lines

```
template<class F> db quad(F f, db a, db b) {
    const int n = 1000;
    db dif = (b-a)/2/n, tot = f(a)+f(b);
    FOR(i,1,2*n) tot += f(a+i*dif)*(i&1?4:2);
    return tot*dif/3;
}
```

IntegrateAdaptive.h

Description: Unused. Fast integration using adaptive Simpson's rule, exact for polynomials of degree up to 5.
Usage: db z, y;
db h(db x) { return x*x + y*y + z*z <= 1; }
db g(db y) { ::y = y; return quad(h, -1, 1); }
db f(db z) { ::z = z; return quad(g, -1, 1); }
db sphereVol = quad(f,-1,1), pi = sphereVol*3/4; 3b316e, 10 lines

```
template<class F> db simpson(F f, db a, db b) {
    db c = (a+b)/2; return (f(a)+4*f(c)+f(b))* (b-a)/6; }
template<class F> db rec(F& f, db a, db b, db eps, db S) {
    db c = (a+b)/2;
    db S1 = simpson(f,a,c), S2 = simpson(f,c,b), T = S1+S2;
    if (abs(T-S)<=15*eps || b-a<1e-10) return T+(T-S)/15;
```

```
    return rec(f,a,c,eps/2,S1)+rec(f,c,b,eps/2,S2);
}
template<class F> db quad(F f, db a, db b, db eps = 1e-8) {
    return rec(f,a,b,eps,simpson(f,a,b)); }
}
```

Simplex.h

Description: Solves a general linear maximization problem: maximize $c^T x$ subject to $Ax \leq b, x \geq 0$. Returns -inf if there is no solution, inf if there are arbitrarily good solutions, or the maximum value of $c^T x$ otherwise. The input vector is set to an optimal x (or in the unbounded case, an arbitrary solution fulfilling the constraints). Numerical stability is not guaranteed. For better performance, define variables such that $x = 0$ is viable.
Usage: vvd A{{1,-1}, {-1,1}, {-1,-2}};
vd b{1,1,-4}, c{-1,-1}, x;
T val = LPSolver(A, b, c).solve(x);
Time: $\mathcal{O}(NM \cdot \#pivots)$, where a pivot may be e.g. an edge relaxation.
 $\mathcal{O}(2^N)$ in the general case.

```
using T = db; // double probably suffices
using vd = V<T>; using vvd = V<vd>;
const T eps = 1e-8, inf = 1/.0;
```

```
#define ltj(X) if (s== -1 || mp(X[j],N[j])<mp(X[s],N[s])) s=j
struct LPSolver {
    int m, n; // # m = constraints, # n = variables
    vi N, B; // N[j] = non-basic variable (j-th column), = 0
    vvd D; // B[j] = basic variable (j-th row)
    LPSolver(const vvd& A, const vd& b, const vd& c) :
        m(sz(b)), n(sz(c)), N(n+1), B(m), D(m+2, vd(n+2)) {
        FOR(i,m) FOR(j,n) D[i][j] = A[i][j];
        FOR(i,m) B[i] = n+1, D[i][n] = -1, D[i][n+1] = b[i];
        // B[i]: basic variable for each constraint
        // D[i][n]: artificial variable for testing feasibility
        FOR(j,n) N[j] = j, D[m][j] = -c[j];
        // D[m] stores negation of objective,
        // which we want to minimize
        N[n] = -1; D[m+1][n] = 1; // to find initial feasible
    } // solution, minimize artificial variable
    void pivot(int r, int s) { // swap B[r] (row)
        T inv = 1/D[r][s]; // with N[r] (column)
        FOR(i,m+2) if (i != r && abs(D[i][s]) > eps) {
            T binv = D[i][s]*inv;
            FOR(j,n+2) if (j != s) D[i][j] -= D[r][j]*binv;
            D[i][s] = -binv;
        }
        D[r][s] = 1; FOR(j,n+2) D[r][j] *= inv; // scale r-th row
        swap(B[r],N[s]);
    }
    bool simplex(int phase) {
        int x = m+phase-1;
        while (1) { // if phase=1, ignore artificial variable
            int s = -1; FOR(j,n+1) if (N[j] != -phase) ltj(D[x]);
            // find most negative col for nonbasic (NB) variable
            if (D[x][s] >= -eps) return 1;
            // can't get better sol by increasing NB variable
            int r = -1;
            FOR(i,m) {
                if (D[i][s] <= eps) continue;
                if (r == -1 || mp(D[i][n+1] / D[i][s], B[i])
                    < mp(D[r][n+1] / D[r][s], B[r])) r = i;
                // find smallest positive ratio
            } // -> max increase in NB variable
            if (r == -1) return 0; // objective is unbounded
            pivot(r,s);
        }
    }
    T solve(vd& x) { // 1. check if x=0 feasible
        int r = 0; FOR(i,1,m) if (D[i][n+1] < D[r][n+1]) r = i;
        if (D[r][n+1] < -eps) { // if not, find feasible start
```

c99f9c, 67 lines

```
    pivot(r,n); // make artificial variable basic
    assert(simplex(2)); // I think this will always be true??
    if (D[m+1][n+1] < -eps) return -inf;
    // D[m+1][n+1] is max possible value of the negation of
    // artificial variable, optimal value should be zero
    // if exists feasible solution
    FOR(i,m) if (B[i] == -1) { // artificial var basic
        int s = 0; FOR(j,1,n+1) ltj(D[i]); // -> nonbasic
        pivot(i,s);
    }
    }
    bool ok = simplex(1); x = vd(n);
    FOR(i,m) if (B[i] < n) x[B[i]] = D[i][n+1];
    return ok ? D[m][n+1] : inf;
}
};
```

Graphs (7)

Erdos-Gallai: $d_1 \geq \dots \geq d_n$ can be degree sequence of simple graph on n vertices iff their sum is even and $\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k), \forall 1 \leq k \leq n$.

7.1 Basics

GridBFS.h

```
Description: BFS through grid with fixed xdir and ydir arrays.
using P = array<int, 2>;
tuple<int, int, char> moves[]{
    {0, 1, 'R'}, {1, 0, 'D'}, {0, -1, 'L'}, {-1, 0, 'U'}
};
struct GridBFS {
    int N, M;
    vector<vector<int>>> dist;
    vector<string> grid, dir;
    vector<vector<P>>> pre;
    GridBFS(int _N, int _M, vector<string>& _g) {
        N = _N, M = _M, grid = _g;
        dist.resize(N, vector(M, -1));
        dir.resize(N, string(M, '.'));
        pre.resize(N, vector<P>(M, {-1, -1}));
    }
    bool valid(int x, int y) {
        return !(x < 0 || x >= N || y < 0 || y >= M);
    }
    void gen(P& st) {
        queue<P> todo;
        todo.push(st); dist[st[0]][st[1]] = 0;
        while (todo.size()) {
            auto u = todo.front(); todo.pop();
            for (auto& [dx, dy, di]: moves) {
                P v = {u[0]+dx, u[1]+dy};
                if (!valid(v[0], v[1]) || dist[v[0]][v[1]] != -1)
                    continue;
                todo.push(v);
                dist[v[0]][v[1]] = dist[u[0]][u[1]] + 1;
                pre[v[0]][v[1]] = u, dir[v[0]][v[1]] = di;
            }
        }
    }
    string getPath(P& st, P en) {
        if (dist[en[0]][en[1]] == -1) return "";
        string res;
        while (en != st) {
            auto& [x, y] = en;
            res += dir[x][y]; en = pre[x][y];
        }
    }
}
```

e9585f, 43 lines

```
    }
    ranges::reverse(res);
    return res;
}
};

Bipartite.h
Description: Checks if an undirected graph is bipartite. Colors nodes with 1/2. If not bipartite, returns false.
Time:  $\mathcal{O}(N + M)$ 
struct Bipartite {
    int N;
    vector<int> color;
    vector<vector<int>>> adj;
    Bipartite(int _N) {
        N = _N;
        adj.resize(N); color.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    bool check() {
        queue<int> todo;
        for (int src = 0; src < N; src++) {
            if (color[src]) continue;
            color[src] = 1; todo.push(src);
            while (!todo.empty()) {
                int u = todo.front(); todo.pop();
                for (auto &v: adj[u]) {
                    if (color[v] == color[u]) return false;
                    if (!color[v]) color[v] = 3-color[u], todo.push(v);
                }
            }
        }
        return true;
    }
};
```

4f7998, 28 lines

TopoSort.h

Description: sorts vertices such that if there exists an edge $u \rightarrow v$, then u goes before v

7b10d8, 20 lines

```
struct TopoSort {
    int N;
    vector<int> in, res;
    vector<vector<int>>> adj;
    TopoSort(int _N) {
        N = _N; in.resize(N); adj.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v), ++in[v];
    }
    bool sort() {
        queue<int> todo;
        for (int u = 0; u < N; ++u) if (!in[u]) todo.push(u);
        while (size(todo)) {
            int u = todo.front(); todo.pop(); res.push_back(u);
            for (auto &v: adj[u]) if (!(--in[v])) todo.push(v);
        }
        return (int)size(res) == N;
    }
};
```

DirectedCycle.h

Description: stack

```
struct DirectedCycle {
```

c746ee, 28 lines


```
int N;
vector<bool> inStk, vis;
vector<int> stk, cyc;
vector<vector<int>> adj;
DirectedCycle(int _N) {
    N = _N;
    adj.resize(N);
    inStk.resize(N), vis.resize(N);
}
void ae(int u, int v) { adj[u].push_back(v); }
vector<int> gen() {
    for (int u = 0; u < N; u++) {
        if (!vis[u] && empty(cyc)) dfs(u);
    }
    return cyc;
}
void dfs(int u) {
    stk.push_back(u);
    inStk[u] = vis[u] = 1;
    for (auto& v: adj[u]) {
        if (inStk[v]) cyc = {ranges::find(stk, v), end(stk)};
        else if (!vis[v]) dfs(v);
        if (size(cyc)) return;
    }
    stk.pop_back(); inStk[u] = 0;
}
};
```

UnDirectedCycle.h

Description: Undirected e4a216, 43 lines

```
struct UnDirectedCycle {
    int N, st, ed;
    vector<bool> vis;
    vector<int> cyc, par;
    vector<vector<int>> adj;
    UnDirectedCycle(int _N) {
        N = _N;
        adj.resize(N);
        par.resize(N, -1), vis.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    vector<int> gen() {
        st = -1;
        for (int u = 0; u < N; u++) {
            if (!vis[u] && dfs(u, -1)) break;
        }
        if (st == -1) return {};
        cyc.push_back(st);
        for (int u = ed; u != st; u = par[u]) {
            cyc.push_back(u);
        }
        cyc.push_back(st);

        return cyc;
    }
    bool dfs(int u, int p = -1) {
        vis[u] = true;
        for (auto& v: adj[u]) {
            if (v == p) continue;
            if (vis[v]) {
                ed = u;
                st = v;
                return true;
            }
        }
        par[v] = u;
    }
};
```

```
if (dfs(v, u)) return true;
}
return false;
}
};
```

Dijkstra.h

Description: shortest path
Time: $\mathcal{O}(N \log N + M)$ 8cfac5, 27 lines

```
template<class T, bool directed> struct Dijkstra {
    int N;
    vector<T> dist;
    vector<vector<pair<int, T>>> adj;
    Dijkstra(int _N) {
        N = _N;
        adj.resize(N);
    }
    void ae(int u, int v, T w) {
        adj[u].push_back({v, w});
        if (!directed) adj[v].push_back({u, w});
    }
    void gen(int st) {
        dist.assign(N, numeric_limits<T>::max());
        priority_queue<pair<T, int>> PQ;
        auto relax = [&](int v, T d) {
            if (dist[v] <= d) return;
            dist[v] = d;
            PQ.push({-dist[v], v});
        }; relax(st, 0);
        while (!PQ.empty()) {
            auto [d, u] = PQ.top(); PQ.pop(); d = -d;
            if (dist[u] < d) continue;
            for (auto& [v, w]: adj[u]) relax(v, w+d);
        }
    }
};
```

NegativeCycle.h

Description: use Bellman-Ford (make sure no underflow) e3180d, 22 lines

```
vector<int> NegativeCycle(int N, vector<array<int, 3>> &ed) {
    vector<int64_t> dist(N);
    vector<int> p(N);
    int x = -1;
    for (int i = 0; i < N; i++) {
        x = -1;
        for (auto &[u, v, w] : ed) {
            if (dist[v] > dist[u]+w) {
                dist[v] = dist[u]+w;
                p[v] = u, x = v;
            }
        }
        if (x == -1) return {};
    }

    for (int i = 0; i < N; i++) x = p[x]; // enter cycle
    vector<int> cyc(x);
    while (p[cyc.back()] != x) cyc.push_back(p[cyc.back()]);
    cyc.push_back(x); // complete cycle
    ranges::reverse(cyc);
    return cyc;
}
```

BellmanFord.h

Description: Shortest Path w/ negative edge weights c2a89c, 33 lines

```
struct BellmanFord {
    int N;
```

```
vector<int64_t> dist;
vector<vector<int>> adj;
vector<array<int, 3>> ed;
BellmanFord(int _N) {
    N = _N;
    adj.resize(N);
    dist.resize(N, INF);
}
void ae(int u, int v, int w) {
    adj[u].push_back(v);
    ed.push_back({u, v, w});
}
void genBad(int u) {
    if (dist[u] == -INF) return;
    dist[u] = -INF;
    for (auto &v : adj[u]) genBad(v);
}
void genDist(int src) {
    dist[src] = 0;
    for (int i = 0; i < N; i++) {
        for (auto &[u, v, w] : ed) {
            if (dist[u] < INF) {
                dist[v] = min(dist[v], dist[u]+w);
            }
        }
    }
    for (auto &[u, v, w] : ed) {
        if (dist[u] < INF && dist[v] > dist[u]+w) genBad(v);
    }
}
};
```

FloydWarshall.h

Description: All-Pairs Shortest Path f36765, 24 lines

```
void floydWarshall(vector<vector<int64_t>>& dist) {
    int N = size(dist);
    for (int i = 0; i < N; i++) dist[i][i] = min<int64_t>(dist[i]
        ↩][i], 0);
    for (int k = 0; k < N; k++) {
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < N; j++) {
                if (dist[i][k] != INT64_MAX && dist[k][j] != INT64_MAX)
                    ↩ {
                        dist[i][j] = min(dist[i][j], dist[i][k]+dist[k][j]);
                    }
            }
        }
    }
    for (int k = 0; k < N; k++) {
        if (dist[k][k] < 0) {
            for (int i = 0; i < N; i++) {
                for (int j = 0; j < N; j++) {
                    if (dist[i][k] != INT64_MAX && dist[k][j] !=
                        ↩INT64_MAX) {
                        dist[i][j] = INT64_MIN;
                    }
                }
            }
        }
    }
}
};
```

7.2 Trees

LCAjump.h

Description: Calculates least common ancestor in tree with verts $0 \dots N-1$ and root R using binary jumping.
Memory: $\mathcal{O}(N \log N)$

Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(\log N)$ query

5c0112, 47 lines

```
struct LCA {
    int N, LOG;
    vector<int> depth;
    vector<vector<int>> par, adj;
    LCA(int _N) {
        N = _N; LOG = 1;
        while ((1<<LOG) < N) ++LOG;
        adj.resize(N);
        depth.resize(N);
        par.resize(N, vector<int>(LOG));
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void gen(int R = 0) { par[R][0] = R; dfs(R); }
    void dfs(int u = 0) {
        for (int j = 1; j < LOG; ++j) {
            par[u][j] = par[par[u][j-1]][j-1];
        }
        for (auto& v: adj[u]) {
            if (v == par[u][0]) continue;
            depth[v] = depth[u]+1;
            par[v][0] = u;
            dfs(v);
        }
    }
    int jump(int x, int d) {
        for (int j = 0; j < LOG; ++j) {
            if ((d>>j) & 1) x = par[x][j];
        }
        return x;
    }
    int lca(int u, int v) {
        if (depth[u] > depth[v]) swap(u, v);
        v = jump(v, depth[v]-depth[u]);
        if (u == v) {return u; }
        for (int j = LOG-1; j >= 0; --j) {
            int U = par[u][j], V = par[v][j];
            if (U != V) u = U, v = V;
        }
        return par[u][0];
    }
    int dist(int u, int v) {
        return depth[u] + depth[v] - 2*depth[lca(u, v)];
    }
};
```

TreeDiameter.h

Description: Calculates longest path in tree. The vertex furthest from 0 must be an endpoint of the diameter.

5ed46f, 34 lines

```
struct TreeDiameter {
    int N, dialen;
    vector<vector<int>> adj;
    vector<int> dist, dia, center, par;
    TreeDiameter(int _N) {
        N = _N; dia = {0, 0};
        par.resie(N);adj.resize(N); dist.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void genDist(int R) { dist[R] = 0; dfs(R); }
    void gen() {
        genDist(0);
        for (int i = 0; i < N; ++i) {
```

```
        if (dist[i] > dist[dia[0]]) dia[0] = i;
    }
    genDist(dia[0]);
    for (int i = 0; i < N; ++i) {
        if (dist[i] > dist[dia[1]]) dia[1] = i;
    }
    diaLen = dist[dia[1]];
    int cen = dia[1];
    for (int i = 0; i < diaLen/2; ++i) cen = par[cen];
    center = {cen};
    if (diaLen&1) center.push_back(par[cen]);
}
void dfs(int u, int p = -1) {
    for (auto& v: adj[u]) if (v != p) {
        dist[v] = dist[u]+1; dfs(v, u);
    }
}
};
```

TreeTour.h

Description: Subtree corresponding to x -> range [st[x],en[x]]

ecb316, 21 lines

```
struct TreeTour {
    int N, timer = -1;
    vector<int> start, end;
    vector<vector<int>> adj;
    EulerTour(int _N) {
        N = _N;
        adj.resize(N);
        start.resize(N), end.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int u, int p = -1) {
        start[u] = ++timer;
        for (auto& v: adj[u]) if (v != p) {
            dfs(v, u);
        }
        end[u] = timer;
    }
};
```

LCArm q.h

Description: Euler Tour LCA. Compress takes a subset S of nodes and computes the minimal subtree that contains all the nodes pairwise LCAs and compressing edges. Returns a list of (par, orig_index) representing a tree rooted at 0. The root points to itself.

Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(1)$ LCA, $\mathcal{O}(|S| \log |S|)$ compress

e5a035, 28 lines

```
struct LCA {
    int N; V<vi> adj;
    vi depth, pos, par, rev; // rev is for compress
    vpi tmp; RMQ<pi> r;
    void init(int _N) { N = _N; adj.rsz(N);
        depth = pos = par = rev = vi(N); }
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void dfs(int x) {
        pos[x] = sz(tmp); tmp.eb(depth[x],x);
        each(y,adj[x]) if (y != par[x]) {
            depth[y] = depth[par[y]=x]+1, dfs(y);
            tmp.eb(depth[x],x); }
    }
    void gen(int R = 0) { par[R] = R; dfs(R); r.init(tmp); }
    int lca(int u, int v){
        u = pos[u], v = pos[v]; if (u > v) swap(u,v);
        return r.query(u,v).s; }
    int dist(int u, int v) {
```

```
        return depth[u]+depth[v]-2*depth[lca(u,v)]; }
    vpi compress(vi S) {
        auto cmp = [&](int a, int b) { return pos[a] < pos[b]; };
        sort(all(S),cmp); R0F(i,sz(S)-1) S.pb(lca(S[i],S[i+1]));
        sort(all(S),cmp); S.erase(unique(all(S)),end(S));
        vpi ret{{0,S[0]}}; F0R(i,sz(S)) rev[S[i]] = i;
        FOR(i,1,sz(S)) ret.eb(rev[lca(S[i-1],S[i])],S[i]);
        return ret;
    }
};
```

HLD.h

Description: Heavy-Light Decomposition, add val to verts and query sum in path/subtree.

Time: any tree path is split into $\mathcal{O}(\log N)$ parts

1802e2, 48 lines

```
template<int SZ, bool VALS_IN_EDGES> struct HLD {
    int N; vi adj[SZ];
    int par[SZ], root[SZ], depth[SZ], sz[SZ], ti;
    int pos[SZ]; vi rpos; // rpos not used but could be useful
    void ae(int x, int y) { adj[x].pb(y), adj[y].pb(x); }
    void dfsSz(int x) {
        sz[x] = 1;
        each(y,adj[x]) {
            par[y] = x; depth[y] = depth[x]+1;
            adj[y].erase(find(all(adj[y]),x));
            dfsSz(y); sz[x] += sz[y];
            if (sz[y] > sz[adj[x][0]]) swap(y,adj[x][0]);
        }
    }
    void dfsHld(int x) {
        pos[x] = ti++; rpos.pb(x);
        each(y,adj[x]) {
            root[y] = (y == adj[x][0] ? root[x] : y);
            dfsHld(y); }
    }
    void init(int _N, int R = 0) { N = _N;
        par[R] = depth[R] = ti = 0; dfsSz(R);
        root[R] = R; dfsHld(R); }
    int lca(int x, int y) {
        for (; root[x] != root[y]; y = par[root[y]])
            if (depth[root[x]] > depth[root[y]]) swap(x,y);
        return depth[x] < depth[y] ? x : y;
    }
    LazySeg<ll,SZ> tree; // segtree for sum
    template <class BinaryOp>
    void processPath(int x, int y, BinaryOp op) {
        for (; root[x] != root[y]; y = par[root[y]]) {
            if (depth[root[x]] > depth[root[y]]) swap(x,y);
            op(pos[root[y]],pos[y]); }
        if (depth[x] > depth[y]) swap(x,y);
        op(pos[x]+VALS_IN_EDGES,pos[y]);
    }
    void modifyPath(int x, int y, int v) {
        processPath(x,y,[this,&v](int l, int r) {
            tree.upd(l,r,v); }); }
    ll queryPath(int x, int y) {
        ll res = 0; processPath(x,y,[this,&res](int l, int r) {
            res += tree.query(l,r); });
        return res; }
    void modifySubtree(int x, int v) {
        tree.upd(pos[x]+VALS_IN_EDGES,pos[x]+sz[x]-1,v); }
};
```

Centroid.h

Description: The centroid of a tree of size N is a vertex such that after removing it, all resulting subtrees have size at most $\frac{N}{2}$. Supports updates in the form “add 1 to all verts v such that $dist(x, v) \leq y$.”

Memory: $\mathcal{O}(N \log N)$

Time: $\mathcal{O}(N \log N)$ build, $\mathcal{O}(\log N)$ update and query

907e21, 54 lines

```
void ad(vi& a, int b) { ckmin(b,sz(a)-1); if (b>=0) a[b]++; }
void prop(vi& a) { R0F(i,sz(a)-1) a[i] += a[i+1]; }
template<int SZ> struct Centroid {
    vi adj[SZ]; void ae(int a,int b){adj[a].pb(b),adj[b].pb(a);}
    bool done[SZ]; // processed as centroid yet
    int N,sub[SZ],cen[SZ],lev[SZ]; // subtree size, centroid anc
    int dist[32-__builtin_clz(SZ)][SZ]; // dists to all ancs
    vi stor[SZ], STOR[SZ];
    void dfs(int x, int p) { sub[x] = 1;
        each(y,adj[x]) if (!done[y] && y != p)
            dfs(y,x), sub[x] += sub[y];
    }
    int centroid(int x) {
        dfs(x,-1);
        for (int sz = sub[x];;) {
            pi mx = {0,0};
            each(y,adj[x]) if (!done[y] && sub[y] < sub[x])
                ckmax(mx,{sub[y],y});
            if (mx.f*2 <= sz) return x;
            x = mx.s;
        }
    }
    void genDist(int x, int p, int lev) {
        dist[lev][x] = dist[lev][p]+1;
        each(y,adj[x]) if (!done[y] && y != p) genDist(y,x,lev); }
    void gen(int CEN, int _x) { // CEN = centroid above x
        int x = centroid(_x); done[x] = 1; cen[x] = CEN;
        sub[x] = sub[_x]; lev[x] = (CEN == -1 ? 0 : lev[CEN]+1);
        dist[lev[x]][x] = 0;
        stor[x].rsz(sub[x]), STOR[x].rsz(sub[x]+1);
        each(y,adj[x]) if (!done[y]) genDist(y,x,lev[x]);
        each(y,adj[x]) if (!done[y]) gen(x,y);
    }
    void init(int _N) { N = _N; FOR(i,1,N+1) done[i] = 0;
        gen(-1,1); } // start at vert 1
    void upd(int x, int y) {
        int cur = x, pre = -1;
        R0F(i,lev[x]+1) {
            ad(stor[cur],y-dist[i][x]);
            if (pre != -1) ad(STOR[pre],y-dist[i][x]);
            if (i > 0) pre = cur, cur = cen[cur];
        }
    } // call propAll() after all updates
    void propAll() { FOR(i,1,N+1) prop(stor[i]), prop(STOR[i]); }
    int query(int x) { // get value at vertex x
        int cur = x, pre = -1, ans = 0;
        R0F(i,lev[x]+1) { // if pre != -1, subtract those from
            ans += stor[cur][dist[i][x]]; // same subtree
            if (pre != -1) ans -= STOR[pre][dist[i][x]];
            if (i > 0) pre = cur, cur = cen[cur];
        }
        return ans;
    }
};
```

7.2.1 SqrtDecompton

HLD generally suffices. If not, here are some common strategies to build the tree after every $\sqrt{N}$ queries.

Centroid DFS SCCT SCCK

- Consider vertices with $>$ or $< \sqrt{N}$ degree separately.
- For subtree updates, note that there are $\mathcal{O}(\sqrt{N})$ distinct sizes among child subtrees of any node.

Block Tree: Use a DFS to split edges into contiguous groups of size $\sqrt{N}$ to $2\sqrt{N}$.

Mo’s Algorithm for Tree Paths: Maintain an array of vertices where each one appears twice, once when a DFS enters the vertex (st) and one when the DFS exists (en). For a tree path $u \leftrightarrow v$ such that $st[u] < st[v]$,

- If u is an ancestor of v , query $[st[u], st[v]]$.
- Otherwise, query $[en[u], st[v]]$ and consider $LCA(u, v)$ separately.

Solutions with worse complexities can be faster if you optimize the operations that are performed most frequently. Use arrays instead of vectors whenever possible. Iterating over an array in order is faster than iterating through the same array in some other order (ex. one given by a random permutation) or DFSing on a tree of the same size. Also, the difference between $\sqrt{N}$ and the optimal block (or buffer) size can be quite large. Try up to $5x$ smaller or larger (at least).

7.3 DFS Algorithms

DFS.h

Description: just DFS

Time: $\mathcal{O}(N + M)$

820647, 19 lines

```
struct DFS {
    int N;
    vector<bool> vis;
    vector<vector<int>>> adj;
    DFS(int _N) {
        N = _N;
        adj.resize(N); vis.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int u) {
        vis[u] = true;
        for (auto& v: adj[u]) {
            if (!vis[v]) dfs(v);
        }
    }
};
```

SCCT.h

Description: Tarjan’s, DFS once to generate strongly connected components in topological order. a, b in same component if both $a \rightarrow b$ and $b \rightarrow a$ exist. Uses less memory than Kosaraju b/c doesn’t store reverse edges.

Time: $\mathcal{O}(N + M)$

7b9c1d, 32 lines

```
struct SCC {
    int N, ti = 0;
    vector<vector<int>>> adj;
    vector<int> disc, comp, stk, comps;
    SCC(int _N) {
        N = _N;
```

```
adj.resize(N);
disc.resize(N), comp.resize(N, -1);
}
void ae(int u, int v) { adj[u].push_back(v); }
int dfs(int u) {
    int low = disc[u] = ++ti;
    stk.push_back(u);
    for (auto& v: adj[u]) if (comp[v] == -1) {
        low = !disc[v] ? min(low, dfs(v)) : min(low, disc[v]);
    }
    if (low == disc[u]) {
        comps.push_back(u);
        for (int v = -1; v != u;) {
            comp[v = stk.back()] = u;
            stk.pop_back();
        }
    }
    return low;
}
void gen() {
    for (int i = 0; i < N; i++) {
        if (!disc[i]) dfs(i);
    }
    ranges::reverse(comps);
}
};
```

SCCK.h

Description: Kosaraju’s Algorithm, DFS twice to generate strongly connected components in topological order. a, b in same component if both $a \rightarrow b$ and $b \rightarrow a$ exist.

Time: $\mathcal{O}(N + M)$

2a0c81, 33 lines

```
struct SCC {
    int N;
    vector<bool> vis;
    vector<vector<int>>> adj, radj;
    vector<int> todo, comp, comps;
    SCC(int _N) {
        N = _N;
        adj.resize(N), radj.resize(N);
        comp.resize(N, -1), vis.resize(N);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        radj[v].push_back(u);
    }
    void dfs(int u) {
        vis[u] = true;
        for (auto &v: adj[u]) if (!vis[v]) dfs(v);
        todo.push_back(u);
    }
    void dfs2(int u, int id) {
        comp[u] = id;
        for (auto &v: radj[u]) if (comp[v] == -1) dfs2(v, id);
    }
    void gen() {
        for (int u = 0; u < N; u++) if (!vis[u]) dfs(u);
        ranges::reverse(todo);
        int id = 0;
        for (auto &u: todo) if (comp[u] == -1) {
            dfs2(u, ++id);
            comps.push_back(u);
        }
    }
};
```

TECC.h

Description: Computes 2-edge-connected components (TECC) of an undirected graph using Tarjan's low-link algorithm. Two vertices belong to the same component iff there is no bridge on any path between them

```
cbdf4f6, 55 lines
struct TECC {
    int N, ti = 0;
    vector<int> disc, low, comp;
    vector<vector<int>>> adj, comps;
    TECC(int _N) {
        N = _N;
        disc.resize(N), low.resize(N);
        adj.resize(N), comp.resize(N, -1);
    }
    void ae(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    bool isBridge(int u, int v) {
        if (disc[u] > disc[v]) swap(u, v);
        return low[v] > disc[u];
    }
    void dfs(int u, int p) {
        bool pSkip = false;
        disc[u] = low[u] = ++ti;
        for (auto& v: adj[u]) {
            if (v == p && !pSkip) {
                pSkip = true;
                continue;
            }
            if (!disc[v]) {
                dfs(v, u);
                low[u] = min(low[u], low[v]);
                // if (isBridge(u, v)) {} // it's a bridge
            }
            else low[u] = min(low[u], disc[v]);
        }
    }
    void getComps(int u) {
        comps[comp[u]].push_back(u);
        for (auto& v: adj[u]) {
            if (~comp[v] || isBridge(u, v)) continue;
            comp[v] = comp[u];
            getComps(v);
        }
    }
    void gen() {
        for (int u = 0; u < N; u++) {
            if (!disc[u]) dfs(u, -1);
        }
        int id = 0;
        for (int u = 0; u < N; u++) {
            if (comp[u] == -1) {
                comp[u] = id++;
                comps.emplace_back();
                getComps(u);
            }
        }
    }
};
```

TwoSAT.h

Description: Calculates a valid assignment to boolean variables a, b, c,... to a 2-SAT problem, so that an expression of the type $(a||b)&&(!a||c)&&(d||!b)&&...$ becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions ($\sim x$).

```
"SCCK.h" 697509, 26 lines
Usage: TwoSAT G(n);
ts.either(0, ~3); // Var 0 is true or var 3 is false
G.gen(N); // Returns true iff it is solvable
G.ans[0..N-1] holds the assigned values to the vars

struct TwoSAT {
    int N;
    vector<bool> ans;
    vector<pair<int, int>> edges;
    TwoSAT(int _N) {
        N = _N;
        ans.resize(N);
    }
    void either(int u, int v) {
        u = max(2*u, -1-2*u), v = max(2*v, -1-2*v);
        edges.push_back({u, v});
    }
    bool gen() {
        SCC S(2*N);
        for (auto& [u, v]: edges) {
            S.ae(u^1, v);
            S.ae(v^1, u);
        }
        S.gen();
        for (int i = 0; i < N; i++) {
            if (S.comp[2*i] == S.comp[2*i^1]) return false;
            ans[i] = S.comp[2*i] > S.comp[2*i^1];
        }
        return true;
    }
};
```

EulerPath.h

Description: Eulerian path starting at src if it exists, visits all edges exactly once. Works for both directed and undirected. Returns vector of {vertex,label of edge to vertex}. Second element of first pair is always -1 .

```
11eb8b, 34 lines
Time:  $\mathcal{O}(N + M)$ 

template<bool directed> struct Euler {
    int N; V<vector<pair<int, int>>> adj;
    V<vector<pair<int, int>>::iterator> its; vector<bool> used;
    Euler(int _N) {
        N = _N;
        adj.resize(N);
    }
    void ae(int u, int v) {
        int M = size(used); used.push_back(0);
        adj[u].push_back({v, M});
        if (!directed) adj[v].push_back({u, M});
    }
    vector<pair<int, int>> gen(int src = 0) {
        its.resize(N);
        for(int i = 0, i < N; i++) its[i] = begin(adj[i]);
        vector<pair<int, int>> ans, s({src, -1}); // {{vert,prev}
        ⇐ vert, edge label}
        int lst = -1; // ans generated in reverse order
        while (size(s)) {
            int x = s.back().first;
            auto& it = its[x], en = end(adj[x]);
            while (it != en && used[it->s]) ++it;
            if (it == en) { // no more edges out of vertex
                if (lst != -1 && lst != x) return {};
                // not a path, no tour exists
                ans.push_back(s.back());
                s.pop_back();
                if (sz(s)) lst = ns.back().first;
            }
            else s.push_back(*it), used[it->s] = 1;
        } // must use all edges
    }
};
```

```
if (size(ans) != sz(used)+1) return {};
ranges::reverse(ans); return ans;
}
};
```

BCC.h

Description: Biconnected components of edges. Removing any vertex in BCC doesn't disconnect it. To get block-cut tree, create a bipartite graph with the original vertices on the left and a vertex for each BCC on the right. Draw edge $u \leftrightarrow v$ if u is contained within the BCC for v . Self-loops are not included in any BCC while BCCs of size 1 represent bridges.

Time: $\mathcal{O}(N + M)$

```
"stdc++.h" ab2730, 45 lines
using namespace std;

struct BCC {
    vector<pair<int, int>> ed;
    vector<vector<pair<int, int>>> adj;
    vector<vector<int>> edgeSets, vertSets; // edges for each bcc
    int N, ti = 0;
    vector<int> disc, stk;
    BCC(int _N) {
        N = _N;
        disc.resize(N), adj.resize(N);
    }
    void ae(int u, int v) {
        adj[u].emplace_back(v, size(ed));
        adj[v].emplace_back(u, size(ed));
        ed.emplace_back(u, v);
    }
    int dfs(int u, int p = -1) { // return lowest disc
        int low = disc[u] = ++ti;
        for (auto e : adj[u]) if (e.s != p) {
            if (!disc[e.f]) {
                stk.pb(e.s); // disc[x] < LOW -> bridge
                int LOW = dfs(e.f, e.s); ckmin(low, LOW);
                if (disc[x] <= LOW) { // get edges in bcc
                    edgeSets.eb(); vi& tmp = edgeSets.bk; // new bcc
                    for (int y = -1; y != e.s; )
                        tmp.pb(y = stk.bk), stk.pop_back();
                }
            }
            else if (disc[e.f] < disc[x]) // back-edge
                ckmin(low, disc[e.f]), stk.pb(e.s);
        }
        return low;
    }
    void gen() {
        F0R(i, N) if (!disc[i]) dfs(i);
        vb in(N);
        each(c, edgeSets) { // edges contained within each BCC
            vertSets.eb(); // so you can easily create block cut tree
            auto ad = [&](int x) {
                if (!in[x]) in[x] = 1, vertSets.bk.pb(x);
            };
            each(e, c) ad(ed[e].f), ad(ed[e].s);
            each(e, c) in[ed[e].f] = in[ed[e].s] = 0;
        }
    }
};
```

MaximalCliques.h

Description: Used only once. Finds all maximal cliques.

Time: $\mathcal{O}(3^{N/3})$

```
f5cd93, 16 lines
using B = bitset<128>; B adj[128];
int N;
// possibly in clique, not in clique, in clique
void cliques(B P = ~B(), B X={}, B R={}) {
    if (!P.any()) {
```

```
    if (!X.any()) // do smth with R
    return;
}
int q = (P[X]._Find_first());
// clique must contain q or non-neighbor of q
B cands = P&~adj[q];
F0R(i,N) if (cands[i]) {
    R[i] = 1; cliques(P&adj[i],X&adj[i],R);
    R[i] = P[i] = 0; X[i] = 1;
}
}
```

7.4 DSU

DSU.h
Description: Disjoint Set Union with path compression and union by size. Add edges and test connectivity. Use for Kruskal’s or Boruvka’s minimum spanning tree.
Time: $\mathcal{O}(\alpha(N))$

```
struct DSU {
    vector<int> e;
    DSU(int N) { e = vector<int>(N, -1); }
    int get(int u) { return e[u] < 0 ? u : e[u] = get(e[u]); }
    bool sameSet(int u, int v) { return get(u) == get(v); }
    int size(int u) { return -e[get(u)]; }
    bool unite(int u, int v) { // union by size
        u = get(u), v = get(v); if (u == v) return false;
        if (e[u] > e[v]) swap(u, v);
        e[u] += e[v]; e[v] = u; return true;
    }
}
```

Kruskal.h
Description: Computes the weight of a Minimum Spanning Tree (MST) using Kruskal’s algorithm. Requires DSU.
Time: $\mathcal{O}(M \log M)$, where M = number of edges

```
"DSU.h"
template <class T> T Kruskal(int N, vector<array<T, 3>> eg) {
    ranges::sort(eg);
    DSU D(N); T ans = 0;
    for (auto &[w, u, v] : eg) if (D.unite(u, v)) ans += w;
    return ans;
}
```

DSUrb.h
Description: Disjoint Set Union with Rollback

```
struct DSUrb {
    vector<int> e;
    vector<array<int,4>> mod;
    DSUrb(int n) { e = vector<int>(n,-1); }
    int get(int x) { return e[x] < 0 ? x : get(e[x]); }
    bool sameSet(int a, int b) { return get(a) == get(b); }
    int size(int x) { return -e[get(x)]; }
    bool unite(int x, int y) { // union-by-rank
        x = get(x), y = get(y);
        if (x == y) {
            mod.push_back({-1,-1,-1,-1});
            return 0;
        }
        if (e[x] > e[y]) swap(x,y);
        mod.push_back({x,y,e[x],e[y]});
        e[x] += e[y]; e[y] = x; return 1;
    }
    void rollback() {
        auto a = mod.back(); mod.pop_back();
        if (a[0] != -1) {
            e[a[0]] = a[2];

```

```
        e[a[1]] = a[3];
    }
};
```

7.5 Flows

Konig’s Theorem: In a bipartite graph, max matching = min vertex cover.

Dilworth’s Theorem: For any partially ordered set, the sizes of the max antichain and of the min chain decomposition are equal. Equivalent to Konig’s theorem on the bipartite graph (U, V, E) where $U = V = S$ and (u, v) is an edge when $u < v$. Those vertices outside the min vertex cover in both U and V form a max antichain.

Dinic.h
Description: Fast flow. After computing flow, edges $\{u, v\}$ such that $lev[u] \neq -1, lev[v] = -1$ are part of min cut. Use reset and rcap for Gomory-Hu.

Time: $\mathcal{O}(N^2M)$ flow, $\mathcal{O}(M\sqrt{N})$ bipartite matching

```
struct Dinic {
    using F = ll; // flow type
    struct Edge { int to; F flo, cap; };
    int N; V<Edge> eds; V<vi> adj;
    void init(int _N) { N = _N; adj.rsz(N), cur.rsz(N); }
    void ae(int u, int v, F cap, F rcap = 0) { assert(min(cap,
        ↪rcap) >= 0);
        adj[u].pb(sz(eds)); eds.pb({v,0,cap});
        adj[v].pb(sz(eds)); eds.pb({u,0,rcap});
    }
    vi lev; V<vi::iterator> cur;
    bool bfs(int s, int t) { // level = shortest distance from
        ↪source
        lev = vi(N,-1); F0R(i,N) cur[i] = begin(adj[i]);
        queue<int> q({s}); lev[s] = 0;
        while (sz(q)) { int u = q.ft; q.pop();
            each(e,adj[u]) { const Edge& E = eds[e];
                int v = E.to; if (lev[v] < 0 && E.flo < E.cap)
                    q.push(v), lev[v] = lev[u]+1;
            }
        }
        return lev[t] >= 0;
    }
    F dfs(int v, int t, F flo) {
        if (v == t) return flo;
        for (; cur[v] != end(adj[v]); cur[v]++) {
            Edge& E = eds[*cur[v]];
            if (lev[E.to]!=lev[v]+1||E.flo==E.cap) continue;
            F df = dfs(E.to,t,min(flo,E.cap-E.flo));
            if (df) { E.flo += df; eds[*cur[v]^1].flo -= df;
                return df; } // saturated >=1 one edge
        }
        return 0;
    }
    F maxFlow(int s, int t) {
        F tot = 0; while (bfs(s,t)) while (F df =
            dfs(s,t,numeric_limits<F>::max())) tot += df;
        return tot;
    }
};
```

GomoryHu.h
Description: Returns edges of Gomory-Hu tree (second element is weight). Max flow between pair of vertices of undirected graph is given by min edge weight along tree path. Uses the fact that for any $i, j, k, \lambda_{ik} \geq \min(\lambda_{ij}, \lambda_{jk})$, where λ_{ij} denotes the flow between i and j .
Time: $N - 1$ calls to Dinic

```
"Dinic.h"
template<class F> V<pair<pi,F>> gomoryHu(int N,
    const V<pair<pi,F>>& ed) {
    vi par(N); Dinic<F> D; D.init(N);
    vpi ed_locs; each(t,ed)ed_locs.pb(D.ae(t.f.f,t.f.s,t.s,t.s));
    V<pair<pi,F>> ans;
    F0R(i,1,N) {
        each(p,ed_locs) { // reset capacities
            auto& e = D.adj.at(p.f).at(p.s);
            auto& e_rev = D.adj.at(e.to).at(e.rev);
            e.cap = e_rev.cap = (e.cap+e_rev.cap)/2;
        }
        ans.pb({{i,par[i]},D.maxFlow(i,par[i])});
        F0R(j,i+1,N) if (par[j] == par[i] && D.lev[j]) par[j] = i;
    }
    return ans;
}
```

MCMF.h
Description: Minimum-cost maximum flow, assumes no negative cycles. It is possible to choose negative edge costs such that the first run of Dijkstra is slow, but this hasn’t been an issue in the past. Edge weights ≥ 0 for every subsequent run. To get flow through original edges, assign ID’s during ae.
Time: Ignoring first run of Dijkstra, $\mathcal{O}(FM \log M)$ if caps are integers and F is max flow.

```
struct MCMF {
    using F = ll; using C = ll; // flow type, cost type
    struct Edge { int to; F flo, cap; C cost; };
    int N; V<C> p, dist; vi pre; V<Edge> eds; V<vi> adj;
    void init(int _N) { N = _N;
        p.rsz(N), dist.rsz(N), pre.rsz(N), adj.rsz(N); }
    void ae(int u, int v, F cap, C cost) { assert(cap >= 0);
        adj[u].pb(sz(eds)); eds.pb({v,0,cap,cost});
        adj[v].pb(sz(eds)); eds.pb({u,0,0,-cost});
    } // use asserts, don't try smth dumb
    bool path(int s, int t) { // find lowest cost path to send
        ↪flow through
        const C inf = numeric_limits<C>::max(); F0R(i,N) dist[i] =
            ↪inf;
        using T = pair<C,int>; priority_queue<T,vector<T>,greater<T
            ↪>> todo;
        todo.push({dist[s] = 0,s});
        while (sz(todo)) { // Dijkstra
            T x = todo.top(); todo.pop(); if (x.f > dist[x.s])
                ↪continue;
            each(e,adj[x.s]) { const Edge& E = eds[e]; // all weights
                ↪ should be non-negative
                if (E.flo < E.cap && ckmin(dist[E.to],x.f+E.cost+p[x.s
                    ↪]-p[E.to]))
                    pre[E.to] = e, todo.push({dist[E.to],E.to});
            }
        } // if costs are doubles, add some EPS so you
        // don't traverse ~0-weight cycle repeatedly
        return dist[t] != inf; // return flow
    }
    pair<F,C> calc(int s, int t) { assert(s != t);
        F0R(_N) F0R(e,sz(eds)) { const Edge& E = eds[e]; //
            ↪Bellman-Ford
            if (E.cap) ckmin(p[E.to],p[eds[e^1].to]+E.cost); }
        F totFlow = 0; C totCost = 0;
        while (path(s,t)) { // p -> potentials for Dijkstra

```

```

F0R(i,N) p[i] += dist[i]; // don't matter for unreachable
    ↪ nodes
F df = numeric_limits<F>::max();
for (int x = t; x != s; x = eds[pre[x]^1].to) {
    const Edge& E = eds[pre[x]]; ckmin(df,E.cap-E.flo); }
totFlow += df; totCost += (p[t]-p[s])*df;
for (int x = t; x != s; x = eds[pre[x]^1].to)
    eds[pre[x]].flo += df, eds[pre[x]^1].flo -= df;
} // get max flow you can send along path
return {totFlow,totCost};
}
};

```

7.6 Matching

Hungarian.h

Description: Given J jobs and W workers ($J \leq W$), computes the minimum cost to assign each prefix of jobs to distinct workers.

@tparam T a type large enough to represent integers on the order of $J * \max(\text{---}C\text{---})$ @param C a matrix of dimensions $J \times W$ such that $C[j][w]$ = cost to assign j-th job to w-th worker (possibly negative)

@return a vector of length J, with the j-th entry equaling the minimum cost to assign the first (j+1) jobs to distinct workers

Time: $\mathcal{O}(J^2W)$

f382f7, 36 lines

```

template <class T> vector<T> hungarian(const vector<vector<T>>
    ↪&C) {
    const int J = (int)size(C), W = (int)size(C[0]);
    assert(J <= W);
    vector<int> job(W + 1, -1);
    vector<T> ys(J), yt(W + 1);
    vector<T> answers;
    const T inf = numeric_limits<T>::max();
    for (int j_cur = 0; j_cur < J; ++j_cur) {
        int w_cur = W;
        job[w_cur] = j_cur;
        vector<T> min_to(W + 1, inf);
        vector<int> prv(W + 1, -1);
        vector<bool> in_Z(W + 1);
        while (job[w_cur] != -1) {
            in_Z[w_cur] = true;
            const int j = job[w_cur];
            T delta = inf;
            int w_next;
            for (int w = 0; w < W; ++w) {
                if (!in_Z[w]) {
                    if (ckmin(min_to[w], C[j][w] - ys[j] - yt[w]))
                        prv[w] = w_cur;
                    if (ckmin(delta, min_to[w])) w_next = w;
                }
            }
            for (int w = 0; w <= W; ++w) {
                if (in_Z[w]) ys[job[w]] += delta, yt[w] -= delta;
                else min_to[w] -= delta;
            }
            w_cur = w_next;
        }
        for (int w; w_cur != -1; w_cur = w) job[w_cur] = job[w =
            ↪prv[w_cur]];
        answers.push_back(-yt[W]);
    }
    return answers;
}

```

GeneralMatchBlossom.h

Description: Variant on Gabow's Impl of Edmond's Blossom Algorithm. General unweighted max matching with 1-based indexing. If $\text{white}[v] = 0$ after $\text{solve}()$ returns, v is part of every max matching.

Time: $\mathcal{O}(NM)$, faster in practice

fd5ce7, 50 lines

```

struct MaxMatching {
    int N; V<vi> adj;
    V<int> mate, first; vb white; vpi label;
    void init(int _N) { N = _N; adj = V<vi>(N+1);
        mate = first = vi(N+1); label = vpi(N+1); white = vb(N+1);
        ↪}
    void ae(int u, int v) { adj.at(u).pb(v), adj.at(v).pb(u); }
    int group(int x) { if (white[first[x]]) first[x] = group(
        ↪first[x]);
        return first[x]; }
    void match(int p, int b) {
        swap(b,mate[p]); if (mate[b] != p) return;
        if (!label[p].s) mate[b] = label[p].f, match(label[p].f,b);
        ↪ // vertex label
        else match(label[p].f,label[p].s), match(label[p].s,label[p]
            ↪.f); // edge label
    }
    bool augment(int st) { assert(st);
        white[st] = 1; first[st] = 0; label[st] = {0,0};
        queue<int> q; q.push(st);
        while (!q.empty()) {
            int a = q.ft; q.pop(); // outer vertex
            each(b,adj[a]) { assert(b);
                if (white[b]) { // two outer vertices, form blossom
                    int x = group(a), y = group(b), lca = 0;
                    while (x||y) {
                        if (y) swap(x,y);
                        if (label[x] == pi{a,b}) { lca = x; break; }
                        label[x] = {a,b}; x = group(label[mate[x]].first);
                    }
                    for (int v: {group(a),group(b)}) while (v != lca) {
                        assert(!white[v]); // make everything along path
                        ↪white
                        q.push(v); white[v] = true; first[v] = lca;
                        v = group(label[mate[v]].first);
                    }
                } else if (!mate[b]) { // found augmenting path
                    mate[b] = a; match(a,b); white = vb(N+1); // reset
                    return true;
                } else if (!white[mate[b]]) {
                    white[mate[b]] = true; first[mate[b]] = b;
                    label[b] = {0,0}; label[mate[b]] = pi{a,0};
                    q.push(mate[b]);
                }
            }
        }
        return false;
    }
    int solve() {
        int ans = 0;
        FOR(st,1,N+1) if (!mate[st]) ans += augment(st);
        FOR(st,1,N+1) if (!mate[st] && !white[st]) assert(!augment(
            ↪st));
        return ans;
    }
};

```

GeneralWeightedMatch.h

Description: General max weight max matching with 1-based indexing. Edge weights must be positive, combo of UnweightedMatch and Hungarian.

Time: $\mathcal{O}(N^3)$?

120873, 145 lines

```

template<int SZ> struct WeightedMatch {
    struct edge { int u,v,w; }; edge g[SZ*2][SZ*2];
    void ae(int u, int v, int w) { g[u][v].w = g[v][u].w = w; }
    int N,NX,lab[SZ*2],match[SZ*2],slack[SZ*2],st[SZ*2];
    int par[SZ*2],floFrom[SZ*2][SZ],S[SZ*2],aux[SZ*2];
    vi flo[SZ*2]; queue<int> q;

```

```

    void init(int _N) { N = _N; // init all edges
        FOR(u,1,N+1) FOR(v,1,N+1) g[u][v] = {u,v,0}; }
    int eDelta(edge e) { // >= 0 at all times
        return lab[e.u]+lab[e.v]-g[e.u][e.v].w*2; }
    void updSlack(int u, int x) { // smallest edge -> blossom x
        if (!slack[x] || eDelta(g[u][x]) < eDelta(g[slack[x]][x]))
            slack[x] = u; }
    void setSlack(int x) {
        slack[x] = 0; FOR(u,1,N+1) if (g[u][x].w > 0
            && st[u] != x && S[st[u]] == 0) updSlack(u,x); }
    void qPush(int x) {
        if (x <= N) q.push(x);
        else each(t,flo[x]) qPush(t); }
    void setSt(int x, int b) {
        st[x] = b; if (x > N) each(t,flo[x]) setSt(t,b); }
    int getPr(int b, int xr) { // get even position of xr
        int pr = find(all(flo[b]),xr)-begin(flo[b]);
        if (pr&1) { reverse(1+all(flo[b])); return sz(flo[b])-pr; }
        return pr; }
    void setMatch(int u, int v) { // rearrange flo[u], matches
        edge e = g[u][v]; match[u] = e.v; if (u <= N) return;
        int xr = floFrom[u][e.u], pr = getPr(u,xr);
        F0R(i,pr) setMatch(flo[u][i],flo[u][i^1]);
        setMatch(xr,v); rotate(begin(flo[u]),pr+all(flo[u])); }
    void augment(int u, int v) { // set matches including u->v
        while (1) { // and previous ones
            int xnv = st[match[u]]; setMatch(u,v);
            if (!xnv) return;
            setMatch(xnv,st[par[xnv]]);
            u = st[par[xnv]], v = xnv;
        }
    }
    int lca(int u, int v) { // same as in unweighted
        static int t = 0; // except maybe return 0
        for (++t;u||v;swap(u,v)) {
            if (!u) continue;
            if (aux[u] == t) return u;
            aux[u] = t; u = st[match[u]];
            if (u) u = st[par[u]];
        }
        return 0;
    }
    void addBlossom(int u, int anc, int v) {
        int b = N+1; while (b <= NX && st[b]) ++b;
        if (b > NX) ++NX; // new blossom
        lab[b] = S[b] = 0; match[b] = match[anc]; flo[b] = {anc};
        auto blossom = [&](int x) {
            for (int y; x != anc; x = st[par[y]])
                flo[b].pb(x), flo[b].pb(y = st[match[x]]), qPush(y);
        };
        blossom(u); reverse(1+all(flo[b])); blossom(v); setSt(b,b);
        // identify all nodes in current blossom
        FOR(x,1,NX+1) g[b][x].w = g[x][b].w = 0;
        FOR(x,1,N+1) floFrom[b][x] = 0;
        each(xs,flo[b]) { // find tightest constraints
            FOR(x,1,NX+1) if (g[b][x].w == 0 || eDelta(g[xs][x]) <
                eDelta(g[b][x])) g[b][x]=g[xs][x], g[x][b]=g[x][xs];
            FOR(x,1,N+1) if (floFrom[xs][x]) floFrom[b][x] = xs;
        } // floFrom to deconstruct blossom
        setSlack(b); // since didn't qPush everything
    }
    void expandBlossom(int b) {
        each(t,flo[b]) setSt(t,t); // undo setSt(b,b)
        int xr = floFrom[b][g[b][par[b]].u], pr = getPr(b,xr);
        for(int i = 0; i < pr; i += 2) {
            int xs = flo[b][i], xns = flo[b][i+1];
            par[xs] = g[xns][xs].u; S[xs] = 1; // no setSlack(xns)?
            S[xns] = slack[xs] = slack[xns] = 0; qPush(xns);
        }
    }

```



```

    S[xr] = 1, par[xr] = par[b];
    FOR(i,pr+1,sz(flo[b])) { // matches don't change
        int xs = flo[b][i]; S[xs] = -1, setSlack(xs); }
    st[b] = 0; // blossom killed
}
bool onFoundEdge(edge e) {
    int u = st[e.u], v = st[e.v];
    if (S[v] == -1) { // v unvisited, matched with smth else
        par[v] = e.u, S[v] = 1; slack[v] = 0;
        int nu = st[match[v]]; S[nu] = slack[nu] = 0; qPush(nu);
    } else if (S[v] == 0) {
        int anc = lca(u,v); // if 0 then match found!
        if (!anc) return augment(u,v), augment(v,u), 1;
        addBlossom(u,anc,v);
    }
    return 0;
}
bool matching() {
    q = queue<int>();
    FOR(x,1,NX+1) {
        S[x] = -1, slack[x] = 0; // all initially unvisited
        if (st[x] == x && !match[x]) par[x] = S[x] = 0, qPush(x);
    }
    if (!sz(q)) return 0;
    while (1) {
        while (sz(q)) { // unweighted matching with tight edges
            int u = q.ft; q.pop(); if (S[st[u]] == 1) continue;
            FOR(v,1,N+1) if (g[u][v].w > 0 && st[u] != st[v]) {
                if (eDelta(g[u][v]) == 0) { // condition is strict
                    if (onFoundEdge(g[u][v])) return 1;
                } else updSlack(u,st[v]);
            }
        }
        int d = INT_MAX;
        FOR(b,N+1,NX+1) if (st[b] == b && S[b] == 1)
            ckmin(d,lab[b]/2); // decrease lab[b]
        FOR(x,1,NX+1) if (st[x] == x && slack[x]) {
            if (S[x] == -1) ckmin(d,eDelta(g[slack[x]][x]));
            else if (S[x] == 0) ckmin(d,eDelta(g[slack[x]][x])/2);
        } // edge weights shouldn't go below 0
        FOR(u,1,N+1) {
            if (S[st[u]] == 0) {
                if (lab[u] <= d) return 0; // why?
                lab[u] -= d;
            } else if (S[st[u]] == 1) lab[u] += d;
        } // lab has opposite meaning for verts and blossoms
        FOR(b,N+1,NX+1) if (st[b] == b && S[b] != -1)
            lab[b] += (S[b] == 0 ? 1 : -1)*d*2;
        q = queue<int>();
        FOR(x,1,NX+1) if (st[x]==x && slack[x] // new tight edge
            && st[slack[x]] != x && eDelta(g[slack[x]][x]) == 0)
            if (onFoundEdge(g[slack[x]][x])) return 1;
        FOR(b,N+1,NX+1) if (st[b]==b && S[b]==1 && lab[b]==0)
            expandBlossom(b); // odd dist blossom taken apart
    }
    return 0;
}
pair<ll,int> calc() {
    NX = N; st[0] = 0; FOR(i,1,2*N+1) aux[i] = 0;
    FOR(i,1,N+1) match[i] = 0, st[i] = i, flo[i].clear();
    int wMax = 0;
    FOR(u,1,N+1) FOR(v,1,N+1)
        floFrom[u][v] = (u == v ? u : 0), ckmax(wMax,g[u][v].w);
    FOR(u,1,N+1) lab[u] = wMax; // start high and decrease
    int num = 0; ll wei = 0; while (matching()) ++num;
    FOR(u,1,N+1) if (match[u] && match[u] < u)
        wei += g[u][match[u]].w; // edges in matching
    return {wei,num};
}

```

```
};
```

MaxMatchFast.h

Description: Fast bipartite matching.

Time: $\mathcal{O}(M\sqrt{N})$

ec6c96, 31 lines

```

vpi maxMatch(int L, int R, const vpi& edges) {
    V<vi> adj = V<vi>(L);
    vi nxt(L,-1), prv(R,-1), lev, ptr;
    F0R(i,sz(edges)) adj.at(edges[i].f).pb(edges[i].s);
    while (true) {
        lev = ptr = vi(L); int max_lev = 0;
        queue<int> q; F0R(i,L) if (nxt[i]==-1) lev[i]=1, q.push(i);
        while (sz(q)) {
            int x = q.ft; q.pop();
            for (int y: adj[x]) {
                int z = prv[y];
                if (z == -1) max_lev = lev[x];
                else if (!lev[z]) lev[z] = lev[x]+1, q.push(z);
            }
            if (max_lev) break;
        }
        if (!max_lev) break;
        F0R(i,L) if (lev[i] > max_lev) lev[i] = 0;
        auto dfs = [&](auto self, int x) -> bool {
            for (;ptr[x] < sz(adj[x]);++ptr[x]) {
                int y = adj[x][ptr[x]], z = prv[y];
                if (z == -1 || (lev[z] == lev[x]+1 && self(self,z)))
                    return nxt[x]=y, prv[y]=x, ptr[x]=sz(adj[x]), 1;
            }
            return 0;
        };
        F0R(i,L) if (nxt[i] == -1) dfs(dfs,i);
    }
    vpi ans; F0R(i,L) if (nxt[i] != -1) ans.pb({i,nxt[i]});
    return ans;
}

```

Geometry (8)

8.1 Primitives

PointShort.h

Description: Use in place of complex<T>.

268ecf, 36 lines

```

using T = db; // or ll
const T EPS = 1e-9; // adjust as needed
using P = pair<T,T>; using vP = V<P>; using Line = pair<P,P>;
int sgn(T a) { return (a>EPS)-(a<-EPS); }
T sq(T a) { return a*a; }

T abs2(P p) { return sq(p.f)+sq(p.s); }
T abs(P p) { return sqrt(abs2(p)); }
T arg(P p) { return atan2(p.s,p.f); }
P conj(P p) { return P(p.f,-p.s); }
P perp(P p) { return P(-p.s,p.f); }
P dir(T ang) { return P(cos(ang),sin(ang)); }

P operator+(P l, P r) { return P(l.f+r.f,l.s+r.s); }
P operator-(P l, P r) { return P(l.f-r.f,l.s-r.s); }
P operator*(P l, T r) { return P(l.f*r,l.s*r); }
P operator/(P l, T r) { return P(l.f/r,l.s/r); }
P operator*(P l, P r) { // complex # multiplication
    return P(l.f*r.f-l.s*r.s,l.s*r.f+l.f*r.s); }
P operator/(P l, P r) { return l*conj(r)/abs2(r); }

P unit(const P& p) { return p/abs(p); }
T dot(const P& a, const P& b) { return a.f*b.f+a.s*b.s; }

```

```

T dot(const P& p, const P& a, const P& b) { return dot(a-p,b-p)
    <=>; }
T cross(const P& a, const P& b) { return a.f*b.s-a.s*b.f; }
T cross(const P& p, const P& a, const P& b) {
    return cross(a-p,b-p); }
P reflect(const P& p, const Line& l) {
    P a = l.f, d = l.s-l.f;
    return a+conj((p-a)/d)*d; }
P foot(const P& p, const Line& l) {
    return (p+reflect(p,l))/(T)2; }
bool onSeg(const P& p, const Line& l) {
    return sgn(cross(l.f,l.s,p)) == 0 && sgn(dot(p,l.f,l.s)) <= 0
    <=>; }
ostream& operator<<(ostream& os, P p) {
    return os << "(" << p.f << ", " << p.s << ")"; }

```

8.2 Polygons

ConvexHull.h

Description: top-bottom convex hull

Time: $\mathcal{O}(N \log N)$

```

"Point.h" 868655, 18 lines
pair<vi,vi> ulHull(const vP& v) {
    vi p(sz(v)), u, l; iota(all(p), 0);
    sort(all(p), [&v](int a, int b) { return v[a] < v[b]; });
    each(i,p) {
        #define ADDP(C, cmp) while (sz(C) > 1 && cross(\
            v[C[sz(C)-2]],v[C.bk],v[i]) cmp 0) C.pop_back(); C.pb(i);
        ADDP(u, >=); ADDP(l, <=);
    }
    return {u,l};
}
vi hullInd(const vP& v) { // returns indices in CCW order
    vi u,l; tie(u,l) = ulHull(v); if (sz(l) <= 1) return l;
    if (v[l[0]] == v[l[1]]) return {0};
    l.insert(end(l),l+rall(u)-1); return l;
}
vP hull(const vP& v) {
    vi w = hullInd(v); vP res; each(t,w) res.pb(v[t]);
    return res; }

```

Strings (9)

9.1 Light

KMP.h

Description: f[i] is length of the longest proper suffix of the *i*-th prefix of s that is a prefix of s

Time: $\mathcal{O}(N)$

993fdd, 20 lines

```

vector<int> KMP(string s) {
    int N = s.size();
    vector<int> f(N + 1); f[0] = -1;
    for (int i = 1; i <= N; i++) {
        for (f[i] = f[i-1]; f[i] != -1 && s[f[i]] != s[i-1];) {
            f[i] = f[f[i]];
        }
        ++f[i];
    }
    return f;
}
vector<int> getOc(string a, string b) { // find occurrences of a
    <=> in b
    vector<int> f = KMP(a + "@" + b), res;
    for (int i = a.size(); i <= b.size(); i++) {
        if (f[i+a.size()+1] == a.size()) {
            res.push_back(i - a.size());
        }
    }
}

```

```
    }
    return res;
}
```

Z.h

Description: f[i] is the max len such that s.substr(0,len) == s.substr(i,len)
Usage: getPrefix("abcab","uwetrabcerabcab");
Time: $\mathcal{O}(N)$

```
vector<int> Zfunction(string s){
    int N = s.size(), L = 1, R = 0; s += '#';
    vector<int> z(N); z[0] = N;
    for (int i = 1; i < N; i++) {
        if (i <= R) z[i] = min(R-i+1, z[i-L]);
        while (s[i+z[i]] == s[z[i]]) ++z[i];
        if (i+z[i]-1 > R) L = i, R = i+z[i]-1;
    }
    return z;
}
vector<int> getPrefix(string a, string b) { // find prefixes of
    ↪ a in b
    vector<int> t = Zfunction(a+b);
    t = vector<int>(size(a)+begin(t), end(t));
    for (auto& x: t) x = min(x, (int)size(a));
    return t;
}
```

Manacher.h

Description: length of largest palindrome centered at each character of string
Usage: ps(manacher("abacaba"))
Time: $\mathcal{O}(N)$ and between every consecutive pair

```
vector<int> Manacher(string _S) {
    string S = "@#";
    for (auto c: _S) S += c, S += "#";
    S.back() = '&#39;';

    vector<int> ans(size(S)-1);
    for (int i = 1, lo = 0, hi = 0; i < size(S)-1; i++) {
        if (i != 1) ans[i] = min(hi-i, ans[hi-i+lo]);
        while (S[i-ans[i]-1] == S[i+ans[i]+1]) ++ans[i];
        if (i+ans[i] > hi) lo = i-ans[i], hi = i+ans[i];
    }
    ans.erase(begin(ans));
    for (int i = 0; i < size(ans); i++) {
        if (i%2 == ans[i]%2) ++ans[i];
    }
    return ans;
}
```

MinRotation.h

Description: minimum cyclic shift
Time: $\mathcal{O}(N)$

```
int minRotation(string s) {
    int a = 0, N = size(s); s += s;
    for (int b = 0; b < N; b++) {
        for (int i = 0; i < N; i++) {
            // a is current best rotation found up to b-1
            if (a+i == b || s[a+i] < s[b+i]) { b += max(0, i-1);
            ↪break; }
            // b to b+i-1 can't be better than a to a+i-1
            if (s[a+i] > s[b+i]) { a = b; break; } // new best found
        }
    }
    return a;
}
```

LyndonFactor.h

Description: A string is "simple" if it is strictly smaller than any of its own nontrivial suffixes. The Lyndon factorization of the string s is a factorization $s = w_1w_2 \dots w_k$ where all strings w_i are simple and $w_1 \geq w_2 \geq \dots \geq w_k$. Min rotation gets min index i such that cyclic shift of s starting at i is minimum.
Time: $\mathcal{O}(N)$

```
vs duval(str s) {
    int N = sz(s); vs factors;
    for (int i = 0; i < N; ) {
        int j = i+1, k = i;
        for (; j < N && s[k] <= s[j]; ++j) {
            if (s[k] < s[j]) k = i;
            else ++k;
        }
        for (; i <= k; i += j-k) factors.pb(s.substr(i, j-k));
    }
    return factors;
}
int minRotation(str s) {
    int N = sz(s); s += s;
    vs d = duval(s); int ind = 0, ans = 0;
    while (ans+sz(d[ind]) < N) ans += sz(d[ind++]);
    while (ind && d[ind] == d[ind-1]) ans -= sz(d[ind--]);
    return ans;
}
```

HashRange.h

Description: Polynomial hash for substrings with two bases. 7f6156, 50 lines

```
using H = array<int, 2>;
H makeH(char c) { return {c, c}; }
mt19937 rng(chrono::steady_clock::now().time_since_epoch().
    ↪count());
uniform_int_distribution<int> UID(0.1*MOD, 0.9*MOD);
const H base = {UID(rng), UID(rng)};
H operator+(H l, H r) {
    for (int i = 0; i < 2; i++) {
        if ((l[i] += r[i]) >= MOD) l[i] -= MOD;
    }
    return l;
}
H operator-(H l, H r) {
    for (int i = 0; i < 2; i++) {
        if ((l[i] -= r[i]) < 0) l[i] += MOD;
    }
    return l;
}
H operator*(H l, H r) {
    for (int i = 0; i < 2; i++) {
        l[i] = (1LL*l[i]*r[i]) % MOD;
    }
    return l;
}
vector<H> pows{{1, 1}};
struct HashRange {
    string S; vector<H> sum{{{}};
    void add(char c) {
        S += c;
        sum.push_back(base*sum.back()+makeH(c));
    }
    void add(string s) {
        for (auto &c : s) add(c);
    }
    void extend(int len) {
        while ((int)pows.size() <= len) {
            pows.push_back(base * pows.back());
        }
    }
}
```

```
H hash(int l, int r) {
    int len = r+l-1; extend(len);
    return sum[r+1] - pows[len] * sum[l];
}
uint64_t hash64(int l, int r) {
    H p = hash(l, r);
    return (uint64_t) p[0]<<32|p[1];
}
// int lcp(HashRange& b) { return first_true([&](int x) {
//     return sum[x] != b.sum[x]; }, 0, min(sz(S), sz(b.S)))-1;
// }
};
```

ReverseBW.h

Description: Used only once. Burrows-Wheeler Transform appends # to a string, sorts the rotations of the string in increasing order, and constructs a new string that contains the last character of each rotation. This function reverses the transform.
Time: $\mathcal{O}(N \log N)$

```
str reverseBW(str t) {
    vi nex(sz(t)); iota(all(nex), 0);
    stable_sort(all(nex), [&t](int a, int b){return t[a]<t[b];});
    str ret; for (int i = nex[0]; i; )
        ret += t[i = nex[i]];
    return ret;
}
```

AhoCorasickFixed.h

Description: Aho-Corasick for fixed alphabet. For each prefix, stores link to max length suffix which is also a prefix.
Time: $\mathcal{O}(N \sum)$

```
template<size_t ASZ> struct ACfixed {
    struct Node { AR<int, ASZ> to; int link; };
    V<Node> d{{{}};
    int add(str s) { // add word
        int v = 0;
        each(C, s) {
            int c = C-'a';
            if (!d[v].to[c]) d[v].to[c] = sz(d), d.eb();
            v = d[v].to[c];
        }
        return v;
    }
    void init() { // generate links
        d[0].link = -1;
        queue<int> q; q.push(0);
        while (sz(q)) {
            int v = q.ft; q.pop();
            F0R(c, ASZ) {
                int u = d[v].to[c]; if (!u) continue;
                d[u].link = d[v].link == -1 ? 0 : d[d[v].link].to[c];
                q.push(u);
            }
            if (v) F0R(c, ASZ) if (!d[v].to[c])
                d[v].to[c] = d[d[v].link].to[c];
        }
    }
};
```

SuffixArray.h

Description: Sort suffixes. First element of sa is sz(S), isa is the inverse of sa, and lcp stores the longest common prefix between every two consecutive elements of sa.
Time: $\mathcal{O}(N \log N)$

```
"RMQ.h"
struct SuffixArray {
    str S; int N; vi sa, isa, lcp;
```

```

void init(str _S) { N = sz(S = _S)+1; genSa(); genLcp(); }
void genSa() { // sa has size sz(S)+1, starts with sz(S)
  sa = isa = vi(N); sa[0] = N-1; iota(1+all(sa),0);
  sort(1+all(sa), [&](int a, int b) { return S[a] < S[b]; });
  FOR(i,1,N) { int a = sa[i-1], b = sa[i];
    isa[b] = i > 1 && S[a] == S[b] ? isa[a] : i; }
  for (int len = 1; len < N; len *= 2) { // currently sorted
    // by first len chars
    vi s(sa), is(isa), pos(N); iota(all(pos),0);
    each(t,s) {int T=t-len;if (T>=0) sa[pos[isa[T]]++] = T;}
    FOR(i,1,N) { int a = sa[i-1], b = sa[i];
      isa[b] = is[a]==is[b]&&is[a+len]==is[b+len]?isa[a]:i; }
    }
  }
void genLcp() { // Kasai's Algo
  lcp = vi(N-1); int h = 0;
  FOR(b,N-1) { int a = sa[isa[b]-1];
    while (a+h < sz(S) && S[a+h] == S[b+h]) ++h;
    lcp[isa[b]-1] = h; if (h) h--; }
  R.init(lcp);
}
RMQ<int> R;
int getLCP(int a, int b) { // lcp of suffixes starting at a,b
  if (a == b) return sz(S)-a;
  int l = isa[a], r = isa[b]; if (l > r) swap(l,r);
  return R.query(l,r-1);
}
};

```

SuffixArrayLinear.h

Description: Linear-time suffix array.

Usage: sa.is(s, 26) // all entries must be in [0, 26)

Time: $\mathcal{O}(N)$, ~100ms for $N = 5 \cdot 10^5$

ed0bb4, 46 lines

```

vi sa_is(const vi& s, int upper) {
  int n = sz(s); if (!n) return {};
  vi sa(n); vb ls(n);
  R0F(i,n-1) ls[i] = s[i] == s[i+1] ? ls[i+1] : s[i] < s[i+1];
  vi sum_l(upper), sum_s(upper);
  FOR(i,n) (ls[i] ? sum_l[s[i]+1] : sum_s[s[i]])++;
  FOR(i,upper) {
    if (i) sum_l[i] += sum_s[i-1];
    sum_s[i] += sum_l[i];
  }
  auto induce = [&](const vi& lms) {
    fill(all(sa),-1);
    vi buf = sum_s;
    for (int d: lms) if (d != n) sa[buf[s[d]]++] = d;
    buf = sum_l; sa[buf[s[n-1]]++] = n-1;
    FOR(i,n) {
      int v = sa[i]-1;
      if (v >= 0 && !ls[v]) sa[buf[s[v]]++] = v;
    }
    buf = sum_l;
    R0F(i,n) {
      int v = sa[i]-1;
      if (v >= 0 && ls[v]) sa[--buf[s[v]+1]] = v;
    }
  };
  vi lms_map(n+1,-1), lms; int m = 0;
  FOR(i,1,n) if (!ls[i-1] && ls[i]) lms_map[i]=m++, lms.pb(i);
  induce(lms); // sorts LMS prefixes
  vi sorted_lms;each(v,sa)if (lms_map[v]!=-1)sorted_lms.pb(v);
  vi rec_s(m); int rec_upper = 0; // smaller subproblem
  FOR(i,1,m) { // compare two lms substrings in sorted order
    int l = sorted_lms[i-1], r = sorted_lms[i];
    int end_l = lms_map[l]+1 < m ? lms[lms_map[l]+1] : n;
    int end_r = lms_map[r]+1 < m ? lms[lms_map[r]+1] : n;
    bool same = 0; // whether lms substrings are same
  }
}

```

```

if (end_l-1 == end_r-r) {
  for (;l < end_l && s[l] == s[r]; ++l,++r);
  if (l != n && s[l] == s[r]) same = 1;
}
rec_s[lms_map[sorted_lms[i]]] = (rec_upper += !same);
}
vi rec_sa = sa_is(rec_s,rec_upper+1);
FOR(i,m) sorted_lms[i] = lms[rec_sa[i]];
induce(sorted_lms); // sorts LMS suffixes
return sa;
}

```

TandemRepeats.h

Description: Find all (i,p) such that $s.substr(i,p) == s.substr(i+p,p)$. No two intervals with the same period intersect or touch.

Usage: solve("aaabababa") // {{0, 1, 1}, {2, 5, 2}}

Time: $\mathcal{O}(N \log N)$

"SuffixArray.h" 661326, 13 lines

```

V<AR<int,3>> solve(str s) {
  int N = sz(s); SuffixArray A,B;
  A.init(s); reverse(all(s)); B.init(s);
  V<AR<int,3>> runs;
  for (int p = 1; 2*p <= N; ++p) { // do in  $\mathcal{O}(N/p)$  for period p
    for (int i = 0, lst = -1; i+p <= N; i += p) {
      int l = i-B.getLCP(N-i-p,N-i), r = i-p+A.getLCP(i,i+p);
      if (l > r || l == lst) continue;
      runs.pb({lst = l,r,p}); // for each i in [l,r],
    } // s.substr(i,p) == s.substr(i+p,p)
  }
  return runs;
}

```

9.2 Heavy

PalTree.h

Description: Used infrequently. Palindromic tree computes number of occurrences of each palindrome within string. ans[i][0] stores min even x such that the prefix $s[1..i]$ can be split into exactly x palindromes, ans[i][1] does the same for odd x .

Time: $\mathcal{O}(N \sum)$ for addChar, $\mathcal{O}(N \log N)$ for updAns

8a7d31, 41 lines

```

struct PalTree {
  static const int ASZ = 26;
  struct node {
    AR<int,ASZ> to = AR<int,ASZ>();
    int len, link, oc = 0; // # occurrences of pal
    int slink = 0, diff = 0;
    AR<int,2> seriesAns;
    node(int _len, int _link) : len(_len), link(_link) {}
  };
  str s = ""; V<AR<int,2>> ans = {{0,MOD}};
  V<node> d = {{0,1},{-1,0}}; // dummy pairs of len 0,-1
  int last = 1;
  int getLink(int v) {
    while (s[sz(s)-d[v].len-2] != s.bk) v = d[v].link;
    return v;
  }
  void updAns() { // serial path has  $\mathcal{O}(\log n)$  vertices
    ans.pb({MOD,MOD});
    for (int v = last; d[v].len > 0; v = d[v].slink) {
      d[v].seriesAns=ans[sz(s)-1-d[d[v].slink].len-d[v].diff];
      if (d[v].diff == d[d[v].link].diff)
        FOR(i,2) ckmin(d[v].seriesAns[i],
          d[d[v].link].seriesAns[i]);
      // start of previous oc of link[v]=start of last oc of v
      FOR(i,2) ckmin(ans.bk[i],d[v].seriesAns[i^1]+1);
    }
  }
}

```

```

void addChar(char C) {
  s += C; int c = C-'a'; last = getLink(last);
  if (!d[last].to[c]) {
    d.eb(d[last].len+2,d[getLink(d[last].link)].to[c]);
    d[last].to[c] = sz(d)-1;
    auto& z = d.bk; z.diff = z.len-d[z.link].len;
    z.slink = z.diff == d[z.link].diff
      ? d[z.link].slink : z.link;
  } // max suf with different dif
  last = d[last].to[c]; ++d[last].oc;
  updAns();
}
void numOc() { R0F(i,2,sz(d)) d[d[i].link].oc += d[i].oc; }
};

```

SuffixAutomaton.h

Description: Used infrequently. Constructs minimal deterministic finite automaton (DFA) that recognizes all suffixes of a string. len corresponds to the maximum length of a string in the equivalence class, pos corresponds to the first ending position of such a string, lnk corresponds to the longest suffix that is in a different class. Suffix links correspond to suffix tree of the reversed string!

Time: $\mathcal{O}(N \log \sum)$

a99c6d, 67 lines

```

struct SuffixAutomaton {
  int N = 1; vi lnk{-1}, len{0}, pos{-1}; // suffix link,
  // max length of state, last pos of first occurrence of state
  V<map<char,int>> nex{1}; V<bool> isClone{0};
  // transitions, cloned -> not terminal state
  V<vi> iLnk; // inverse links
  int add(int p, char c) { // ~p nonzero if p != -1
    auto getNext = [&]() {
      if (p == -1) return 0;
      int q = nex[p][c]; if (len[p]+1 == len[q]) return q;
      int clone = N++; lnk.pb(lnk[q]); lnk[q] = clone;
      len.pb(len[p]+1), nex.pb(nex[q]),
      pos.pb(pos[q]), isClone.pb(1);
      for (; ~p && nex[p][c] == q; p = lnk[p]) nex[p][c]=clone;
      return clone;
    };
    // if (nex[p].count(c)) return getNext();
    // ^ need if adding > 1 string
    int cur = N++; // make new state
    lnk.eb(), len.pb(len[p]+1), nex.eb(),
    pos.pb(pos[p]+1), isClone.pb(0);
    for (; ~p && !nex[p].count(c); p = lnk[p]) nex[p][c] = cur;
    int x = getNext(); lnk[cur] = x; return cur;
  }
  void init(str s) { int p = 0; each(x,s) p = add(p,x); }
  // inverse links
  void genILnk() { iLnk.rsz(N);FOR(v,1,N)iLnk[lnk[v]].pb(v); }
  // APPLICATIONS
  void getAllOccur(vi& oc, int v) {
    if (!isClone[v]) oc.pb(pos[v]); // terminal position
    each(u,iLnk[v]) getAllOccur(oc,u); }
  vi allOccur(str s) { // get all occurrences of s in automaton
    int cur = 0;
    each(x,s) {
      if (!nex[cur].count(x)) return {};
      cur = nex[cur][x]; }
    // convert end pos -> start pos
    vi oc; getAllOccur(oc,cur); each(t,oc) t += 1-sz(s);
    sort(all(oc)); return oc;
  }
  vl distinct;
  ll getDistinct(int x) {
    // # distinct strings starting at state x
    if (distinct[x]) return distinct[x];
    distinct[x]=1;each(y,nex[x]) distinct[x]+=getDistinct(y.s);
  }
}

```

```
return distinct[x]; }
ll numDistinct() { // # distinct substrings including empty
distinct.rsz(N); return getDistinct(0); }
ll numDistinct2() { // assert(numDistinct()==numDistinct2());
ll ans = 1; FOR(i,1,N) ans += len[i]-len[lnk[i]];
return ans; }
};

SuffixAutomaton S;
vi sa; str s;
void dfs(int x) {
if (!S.isClone[x]) sa.pb(sz(s)-1-S.pos[x]);
V<pair<char,int>> chr;
each(t,S.iLnk[x]) chr.pb({s[S.pos[t]-S.len[x]],t});
sort(all(chr)); each(t,chr) dfs(t.s);
}

int main() {
re(s); reverse(all(s));
S.init(s); S.genILnk();
dfs(0); ps(sa); // generating suffix array for s
}
```

SuffixTree.h
Description: Used infrequently. Ukkonen’s algorithm for suffix tree. Longest non-unique suffix of s has length len[p]+lef after each call to add terminates. Each iteration of loop within add decreases this quantity by one.
Time: $\mathcal{O}(N \log \Sigma)$

```
struct SuffixTree {
str s; int N = 0;
vi pos, len, lnk; V<map<char,int>> to;
int make(int POS, int LEN) { // lnk[x] is meaningful when
// x!=0 and len[x] != MOD
pos.pb(POS); len.pb(LEN); lnk.pb(-1); to.eb(); return N++; }
void add(int& p, int& lef, char c) { // longest
// non-unique suffix is at node p with lef extra chars
s += c; ++lef; int lst = 0;
for (; lef; p=lnk[p]; lef--) { // if p != root then lnk[p]
// must be defined
while (lef>1 && lef>len[to[p][s[sz(s)-lef]]])
p = to[p][s[sz(s)-lef]]; lef -= len[p];
// traverse edges of suffix tree while you can
char e = s[sz(s)-lef]; int& q = to[p][e];
// next edge of suffix tree
if (!q) q = make(sz(s)-lef,MOD), lnk[lst] = p, lst = 0;
// make new edge
else {
char t = s[pos[q]+lef-1];
if (t == c) { lnk[lst] = p; return; } // suffix not
↳unique
int u = make(pos[q],lef-1);
// new node for current suffix-1, define its link
to[u][c] = make(sz(s)-1,MOD); to[u][t] = q;
// new, old nodes
pos[q] += lef-1; if (len[q] != MOD) len[q] -= lef-1;
q = u, lnk[lst] = u, lst = u;
}
}
}
void init(str _s) {
make(-1,0); int p = 0, lef = 0;
each(c,_s) add(p,lef,c);
add(p,lef,'$'); s.pop_back(); // terminal char
}
int maxPre(str x) { // max prefix of x which is substring
for (int p = 0, ind = 0;;) {
if (ind == sz(x) || !to[p].count(x[ind])) return ind;
p = to[p][x[ind]];
}
```

```
FOR(i,len[p]) {
if (ind == sz(x) || x[ind] != s[pos[p]+i]) return ind;
ind ++;
}
}
}
vi sa; // generate suffix array
void genSa(int x = 0, int Len = 0) {
if (!sz(to[x])) sa.pb(pos[x]-Len); // found terminal node
else each(t,to[x]) genSa(t.s,Len+len[x]);
}
};
```

Various (10)

10.1 Dynamic programming

When doing DP on intervals:
 $a[i][j] = \min_{i < k < j} (a[i][k] + a[k][j]) + f(i, j)$, where the (minimal) optimal k increases with both i and j ,

- one can solve intervals in increasing order of length, and search $k = p[i][j]$ for $a[i][j]$ only between $p[i][j - 1]$ and $p[i + 1][j]$.
- This is known as Knuth DP. Sufficient criteria for this are if $f(b, c) \leq f(a, d)$ and $f(a, c) + f(b, d) \leq f(a, d) + f(b, c)$ for all $a \leq b \leq c \leq d$.
- Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.

CircularLCS.h
Description: Used only twice. For strs A, B calculates longest common subsequence of A with all rotations of B
Time: $\mathcal{O}(|A| \cdot |B|)$

```
int circular_lcs(str A, str B) {
B += B;
int max_lcs = 0;
V<vb> dif_left(sz(A)+1,vb(sz(B)+1)), dif_up(sz(A)+1,vb(sz(B)
↳+1));
auto recalc = [&](int x, int y) { assert(x && y);
int res = (A.at(x-1) == B.at(y-1)) |
dif_up[x][y-1] | dif_left[x-1][y];
dif_left[x][y] = res-dif_up[x][y-1];
dif_up[x][y] = res-dif_left[x-1][y];
};
FOR(i,1,sz(A)+1) FOR(j,1,sz(B)+1) recalc(i,j);
FOR(j,sz(B)/2) {
// 1. zero out dp[.][j], update dif_left and dif_right
if (j) for (int x = 1, y = j; x <= sz(A) && y <= sz(B); ) {
int pre_up = dif_up[x][y];
if (y == j) dif_up[x][y] = 0;
else recalc(x,y);
(pre_up == dif_up[x][y]) ? ++x : ++y;
}
// 2. calculate LCS(A[0:sz(A)),B[j:j+sz(B)/2])
int cur_lcs = 0;
FOR(x,1,sz(A)+1) cur_lcs += dif_up[x][j+sz(B)/2];
ckmax(max_lcs,cur_lcs);
}
return max_lcs;
}
```

SMAWK.h
Description: Given negation of totally monotone matrix with entries of type D , find indices of row maxima (their indices increase for every submatrix). If tie, take lesser index. f returns matrix entry at (r, c) in $\mathcal{O}(1)$. Use in place of divide & conquer to remove a log factor.
Time: $\mathcal{O}(R + C)$, can be reduced to $\mathcal{O}(C(1 + \log R/C))$ evaluations of f

```
template<class F, class D=ll> vi smawk (F f, vi x, vi y) {
vi ans(sz(x),-1); // x = rows, y = cols
#define upd() if (ans[i] == -1 || w > mx) ans[i] = c, mx = w
if (min(sz(x),sz(y)) <= 8) {
FOR(i,sz(x)) { int r = x[i]; D mx;
each(c,y) { D w = f(r,c); upd(); } }
return ans;
}
if (sz(x) < sz(y)) { // reduce subset of cols to consider
vi Y; each(c,y) {
for (;sz(Y);Y.pop_back()) { int X = x[sz(Y)-1];
if (f(X,Y.bk) >= f(X,c)) break; }
if (sz(Y) < sz(x)) Y.pb(c);
} y = Y;
} // recurse on half the rows
vi X; for (int i = 1; i < sz(x); i += 2) X.pb(x[i]);
vi ANS = smawk(f,X,y); FOR(i,sz(ANS)) ans[2*i+1] = ANS[i];
for (int i = 0, k = 0; i < sz(x); i += 2) {
int to = i+1 < sz(ans) ? ans[i+1] : y.bk; D mx;
for(int r = x[i]; ++k) {
int c = y[k]; D w = f(r,c); upd();
if (c == to) break; }
}
return ans;
};
```

10.2 Debugging tricks

- signal(SIGSEGV, [](int) { _Exit(0); }); converts segfaults into Wrong Answers. Similarly one can catch SIGABRT (assertion failures) and SIGFPE (zero divisions). `_GLIBCXX_DEBUG` violations generate SIGABRT (or SIGSEGV on gcc 5.4.0 apparently).
- feenableexcept(29); kills the program on NaNs (1), 0-divs (4), infinities (8) and denormals (16).

10.3 Optimization tricks

10.3.1 Bit hacks

- $x \ \& \ -x$ is the least bit in x .
- for (int x = m; x;) { --x &= m; ... } loops over all subset masks of m (except m itself).
- $c = x \& -x$, $r = x + c$; $((r \wedge x) \gg 2) / c$ | r is the next number after x with the same number of bits set.
- FOR(b,k) FOR(i,1<<K) if (i&1<<b) D[i] += D[i^(1<<b)]; computes all sums of subsets.

10.3.2 STL Utilities

- template<class T> using priority_queue_min = priority_queue<T, vector<T>, greater<T>>; Min-heap priority queue.

- `ranges::rotate(p, ranges::min_element(p));`
Lexicographically smallest cyclic shift.
- `next_permutation(v.begin(), v.end());` Next permutation.
- `prev_permutation(v.begin(), v.end());` Previous permutation.
- `auto [mn, mx] = ranges::minmax(v);` Min and max in one pass.
- `ranges::nth_element(v, v.begin()+k);` k-th smallest in $O(n)$.
- `ranges::binary_search(v, x);` Check existence in sorted vector.
- `ranges::lower_bound(v, x);` First element $\geq x$.
- `ranges::upper_bound(v, x);` First element $> x$.
- `int cnt = ranges::upper_bound(v, R) - ranges::lower_bound(↔v, L);` Count of elements in range $[L, R]$.
- `auto [l,r] = equal_range(v.begin(),v.end(),x);` Range of equal values.
- `v.erase(unique(v.begin(),v.end()),v.end());` Remove duplicates (after sort).
- `iota(v.begin(),v.end(),0);` Fill with increasing values.

10.3.3 Pragmas

- `#pragma GCC optimize ("Ofast")` will make GCC auto-vectorize for loops and optimizes floating points better (assumes associativity and turns off denormals).
- `#pragma GCC target ("avx,avx2")` can double performance of vectorized code, but causes crashes on old machines. Also consider older `#pragma GCC target ("sse4")`.
- `#pragma GCC optimize ("trapv")` kills the program on integer overflows (but is really slow).

FastIO.h
Description: Fast input and output for integers and strings. For doubles, read them as strings and convert them to double using stod.
Usage: `initO(); int a,b; ri(a,b); wi(b,'\n');` `wi(a,'\n');`
Time: input is ~300ms faster for 10^6 long longs on CF. cbf279, 39 lines

```
inline namespace FastIO {

const int BSZ = 1<<15; //INPUT
char ibuf[BSZ]; int ipos, ilen;
char nc() { // next char
    if (ipos == ilen) {
        ipos = 0; ilen = fread(ibuf,1,BSZ,stdin);
        if (!ilen) return EOF;
    }
    return ibuf[ipos++];
}

void rs(str& x) { // read str
    char ch; while (isspace(ch = nc()));
    do { x += ch; } while (!isspace(ch = nc()) && ch != EOF);
}
```

```
}
tcT> void ri(T& x) { // read int or ll
    char ch; int sgn = 1;
    while (!isdigit(ch = nc())) if (ch == '-' ) sgn *= -1;
    x = ch-'0'; while (isdigit(ch = nc())) x = x*10+(ch-'0');
    x *= sgn;
}

tcT, class... Ts> void ri(T& t, Ts&... ts) {
    ri(t); ri(ts...); } // read ints
///// OUTPUT (call initO() at start)
char obuf[BSZ], numBuf[100]; int opos;
void flushOut() { fwrite(obuf,1,opos,stdout); opos = 0; }
void wc(char c) { // write char
    if (opos == BSZ) flushOut();
    obuf[opos++] = c; }
void ws(str s) { each(c,s) wc(c); } // write str
tcT> void wi(T x, char after = '\0') {
    if (x < 0) wc('-'), x *= -1;
    int len = 0; for (;x>=10;x/=10) numBuf[len++] = '0'+(x%10);
    wc('0'+x); R0F(i,len) wc(numBuf[i]);
    if (after) wc(after);
}

void initO() { assert(atexit(flushOut) == 0); }

}
```

10.4 Other languages

Python3.py
Description: Python review. 40 lines

```
from math import *
import sys, random
def nextInt():
    return int(input())
def nextStrs():
    return input().split()
def nextInts():
    return list(map(int,nextStrs()))

n = nextInt()
v = [n]
def process(x):
    global v
    x = abs(x)
    V = []
    for t in v:
        g = gcd(t,x)
        if g != 1:
            V.append(g)
        if g != t:
            V.append(t//g)
    v = V
for i in range(50):
    x = random.randint(0,n-1)
    if gcd(x,n) != 1:
        process(x)
    else:
        sx = x*x%n # assert(gcd(sx,n) == 1)
        print(f"sqrt {sx}")
        sys.stdout.flush()
        X = nextInt()
        process(x+X)
        process(x-X)
print(f'! {len(v)}',end='')
for i in v:
    print(f' {i}',end='')
print()
sys.stdout.flush() # sys.exit(0) -> exit
# sys.setrecursionlimit(int(1e9)) -> stack size
```

```
# print(f'{ans:=.6f}') -> print ans to 6 decimal places

Decimal.py
Description: Arbitrary-precision decimals 5 lines

from decimal import *

getcontext().prec = 100 # how many digits of precision
print(Decimal(1) / Decimal(7)) # 0.142857142857...
print(Decimal(10) ** -100) # 1E-100

10.5 Miscellaneous

GridDirection.h
Description: Predefined direction for grid-based problems. Includes common movement sets for 2D and 3D grids such as 4-directional, 8-directional, diagonal-only, knight moves, 3D etc. efed5e, 22 lines

// Down, Right, Up, Left
const int dx[4] = {1, 0, -1, 0};
const int dy[4] = {0, 1, 0, -1};
const char dc[4] = {'D', 'R', 'U', 'L'};

// Down, Down-Right, Right, Up-Right, Up, Up-Left, Left, Down-Left
const int dx8[8] = {1, 1, 0, -1, -1, -1, 0, 1};
const int dy8[8] = {0, 1, 1, 1, 0, -1, -1, -1};

// Diagonals only
const int dxDiag[4] = {1, 1, -1, -1};
const int dyDiag[4] = {1, -1, 1, -1};

// Knight moves
const int dxK[8] = {2, 2, 1, 1, -1, -1, -2, -2};
const int dyK[8] = {1, -1, 2, -2, 2, -2, 1, -1};

// 3D Grid
const int dx[6] = {1, -1, 0, 0, 0, 0};
const int dy[6] = {0, 0, 1, -1, 0, 0};
const int dz[6] = {0, 0, 0, 0, 1, -1};
```